Resampling Techniques and Bayesian Computation

Subhasish Basak

Outline I

Outline II

Course Overview

This course introduces the main ideas behind resampling and Bayesian methods in modern statistics. You will learn how techniques like the jackknife and bootstrap can be used to measure uncertainty and improve estimates in practical settings.
The course then moves to the Bayesian approach, covering priors, posteriors, credible intervals, and simple computation tools. It wraps up with an introduction to MCMC methods such as Metropolis–Hastings and Gibbs sampling.

Course Overview

Grade distribution:

- Assignments: 20 %
- In-Semester Exam I 20%
- In-Semester Exam II 20%
- End-Semester Exam 40%

Reference Books

- **Resampling:** An Introduction to the Bootstrap — Efron and Tibshirani.
- **Bayesian Analysis:** Bayesian Data Analysis — Gelman et al.
- **MCMC:** Monte Carlo Statistical Methods — Robert and Casella.

Course Overview

In many modern statistical problems,

- Exact sampling distributions are unknown or hard to derive
- Classical assumptions (normality, large sample size) may fail
- limited data
- Can be too computationally expensive

This course studies **resampling techniques** such as bootstrap and jackknife that use data-driven simulation to quantify uncertainty.

Population and Sample

- **Population**: entire set of interest
- **Sample**: subset drawn from population

We make inference about the population using samples.
Inference is typically based on a **statistic** computed from the sample, which is used to draw conclusions about the population.
Assume:

- Each observation is drawn randomly
- Observations are independent

$$X_1, X_2, \ldots, X_n \text{ i.i.d.}$$

Statistics

A **statistic** is any function of the data.
Examples:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

$$X_{(n)} = \max_i X_i$$

Due to to the randomness of the drawn samples statistics are random variables.

Expectation and Variance

Let X be a random variable.

Expectation (Mean)

$$\mathbb{E}[X] = \begin{cases} \sum_x x\,\mathbb{P}(X = x), & \text{(discrete)} \\ \int x f_X(x)\,dx, & \text{(continuous)} \end{cases}$$

Variance

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

- Expectation measures central tendency
- Variance measures spread or uncertainty

Example of Sampling distribution of a statistic using Bernoulli Distribution

A Bernoulli random variable:

$$X = \begin{cases} 1 & \textit{with prob } p \\ 0 & \textit{with prob } 1-p \end{cases}$$

$$\mathbb{E}[X] = p$$

$$\mathrm{Var}(X) = p(1-p)$$

Used to model:

- Yes / No outcomes
- Success / Failure experiments
- coin toss

Statistic for Bernoulli Data

Let

$$X_1, \ldots, X_n \sim \text{Bernoulli}(p)$$

Sample proportion:

$$\hat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{S_n}{n}, \quad \text{where } S_n = \sum_{i=1}^{n} X_i$$

- $S_n \sim \text{Binomial}(n, p)$
- $\hat{p}$ is a scaled Binomial random variable

Exact distribution of $\hat{p}$:

$$\mathbb{P}\left(\hat{p} = \frac{k}{n}\right) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n$$

Sampling Distribution of Sample Proportion

For Bernoulli data, the sampling distribution of the statistic $\hat{p}$ is **known exactly** for any sample size n.

Let

$$S_n = \sum_{i=1}^{n} X_i \sim \text{Binomial}(n, p), \quad \hat{p} = \frac{S_n}{n}$$

From properties of the Binomial distribution:

$$\mathbb{E}[S_n] = np, \qquad \mathrm{Var}(S_n) = np(1-p)$$

Hence,

$$\mathbb{E}[\hat{p}] = p, \qquad \mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

- The exact sampling distribution of $\hat{p}$ is fully known
- No approximation is required in this case

Example of Sampling Distribution using Binomial Data

A Binomial random variable:

$$Y \sim \text{Binomial}(n, p)$$

$$\mathbb{P}(Y = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n$$

$$\mathbb{E}[Y] = np, \qquad \mathrm{Var}(Y) = np(1-p)$$

Used to model:

- Number of successes in n trials
- Aggregated Bernoulli outcomes
- Repeated experiments with fixed n

Statistic for Binomial Data

Let

$$Y_1, \ldots, Y_m \sim \text{Binomial}(n, p)$$

Sample mean:

$$\bar{Y} = \frac{1}{m} \sum_{i=1}^{m} Y_i$$

- $\bar{Y}$ is the average number of successes
- $\sum_{i=1}^{m} Y_i \sim \text{Binomial}(mn, p)$

Exact distribution of $\bar{Y}$:

$$\mathbb{P}\left(\bar{Y} = \frac{k}{m}\right) = \binom{mn}{k} p^k (1-p)^{mn-k}, \quad k = 0, 1, \ldots, mn$$

Sampling Distribution of Sample Mean

For Binomial data, the sampling distribution of the statistic $\bar{Y}$ is **known exactly**.

Let

$$Y_1, \ldots, Y_m \sim \text{Binomial}(n, p), \qquad \bar{Y} = \frac{1}{m}\sum_{i=1}^{m} Y_i$$

From properties of the Binomial distribution:

$$\mathbb{E}[\bar{Y}] = np, \qquad \mathrm{Var}(\bar{Y}) = \frac{np(1-p)}{m}$$

- Exact sampling distribution is discrete
- Variance shrinks as sample size m increases

Example of Sampling Distribution using Poisson Data

A Poisson random variable:

$$Y \sim \text{Poisson}(\lambda)$$

$$\mathbb{P}(Y = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots$$

$$\mathbb{E}[Y] = \lambda, \qquad \mathrm{Var}(Y) = \lambda$$

Used to model:

- Number of events in a fixed time/space interval
- Arrival processes (calls, packets, failures)
- Rare event counts

Statistic for Poisson Data

Let

$$Y_1, \ldots, Y_m \sim \text{Poisson}(\lambda)$$

Sample mean:

$$\bar{Y} = \frac{1}{m} \sum_{i=1}^{m} Y_i$$

- $\bar{Y}$ estimates the rate parameter λ
- $\sum_{i=1}^{m} Y_i \sim \text{Poisson}(m\lambda)$

Exact distribution of $\bar{Y}$:

$$\mathbb{P}\left(\bar{Y} = \frac{k}{m}\right) = \frac{e^{-m\lambda}(m\lambda)^k}{k!}, \quad k = 0, 1, 2, \ldots$$

Sampling Distribution of Sample Mean

For Poisson data, the sampling distribution of the statistic $\bar{Y}$ is **known exactly**.

Let

$$Y_1, \ldots, Y_m \sim \text{Poisson}(\lambda), \quad \bar{Y} = \frac{1}{m}\sum_{i=1}^{m} Y_i$$

From properties of the Poisson distribution:

$$\mathbb{E}[\bar{Y}] = \lambda, \qquad \mathrm{Var}(\bar{Y}) = \frac{\lambda}{m}$$

- $\bar{Y}$ is an unbiased estimator of λ
- Variance decreases as sample size m increases

Example of Sampling Distribution using Poisson Data

A Poisson random variable:

$$Y \sim \text{Poisson}(\lambda)$$

$$\mathbb{P}(Y = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots$$

$$\mathbb{E}[Y] = \lambda, \qquad \mathrm{Var}(Y) = \lambda$$

Used to model:

- Number of events in a fixed interval
- Arrival processes (calls, packets, failures)
- Rare event counts

Statistic for Poisson Data

Let

$$Y_1, \ldots, Y_m \sim \text{Poisson}(\lambda)$$

Sample mean:

$$\bar{Y} = \frac{1}{m} \sum_{i=1}^{m} Y_i$$

- $\bar{Y}$ is a statistic used to estimate λ
- Distribution of $\bar{Y}$ depends on the sum $\sum Y_i$

Convolution of Two Poisson Variables: Setup

Let

$$Y_1 \sim \text{Poisson}(\lambda_1), \quad Y_2 \sim \text{Poisson}(\lambda_2)$$

be independent.

We want the distribution of

$$S = Y_1 + Y_2$$

For $k = 0, 1, 2, \ldots,$

$$\mathbb{P}(S = k) = \mathbb{P}(Y_1 + Y_2 = k)$$

The event $\{Y_1 + Y_2 = k\}$ occurs when:

$$(Y_1, Y_2) = (0, k), (1, k-1), \ldots, (k, 0)$$

Convolution of Two Poisson Variables: Calculation

Using independence,

$$\mathbb{P}(Y_1 + Y_2 = k) = \sum_{j=0}^{k} \mathbb{P}(Y_1 = j)\mathbb{P}(Y_2 = k - j)$$

Substitute Poisson pmfs:

$$= \sum_{j=0}^{k} \frac{e^{-\lambda_1}\lambda_1^j}{j!} \frac{e^{-\lambda_2}\lambda_2^{k-j}}{(k-j)!}$$

Simplifying,

$$= \frac{e^{-(\lambda_1+\lambda_2)}}{k!} \sum_{j=0}^{k} \binom{k}{j} \lambda_1^j \lambda_2^{k-j}$$

By the Binomial theorem,

$$\mathbb{P}(Y_1 + Y_2 = k) = \frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1 + \lambda_2)^k}{k!}$$

Sampling Distribution of Sample Mean

Using the convolution result,

$$\sum_{i=1}^{m} Y_i \sim \text{Poisson}(m\lambda)$$

Therefore,

$$\bar{Y} = \frac{1}{m}\sum_{i=1}^{m} Y_i$$

From properties of the Poisson distribution:

$$\mathbb{E}[\bar{Y}] = \lambda, \qquad \mathrm{Var}(\bar{Y}) = \frac{\lambda}{m}$$

- $\bar{Y}$ is an unbiased estimator of λ
- Variability decreases as sample size m increases

From Discrete to Continuous Distributions

So far, we have worked with **discrete random variables**:

- Bernoulli, Binomial, Poisson
- Take **countable values** $(0,1,2,\ldots)$
- Probabilities assigned to individual values

In many real-life problems:

- Time, distance, voltage, noise, error
- Can take **any value in an interval**

These are modeled using **continuous random variables**.

Continuous Random Variables

A random variable X is said to be **continuous** if:

- It can take any value in an interval of real numbers
- Probabilities are described using a **density function**

Probability density function (pdf):

$$f_X(x) \geq 0, \qquad \int_{-\infty}^{\infty} f_X(x)\,dx = 1$$

Probabilities are computed as:

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x)\,dx$$

Uniform Distribution: A Simple Continuous Model

A continuous random variable X is said to follow a **Uniform**$(0, \theta)$ distribution if its pdf is

$$f_X(x) = \begin{cases} \frac{1}{\theta}, & 0 \leq x \leq \theta \\ 0, & \text{otherwise} \end{cases}$$

- All values between 0 and θ are equally likely
- Probability depends only on the **length of the interval**

$$\mathbb{E}[X] = \frac{\theta}{2}, \qquad \mathrm{Var}(X) = \frac{\theta^2}{12}$$

Probability as Area: Uniform$(0, \theta)$

For $0 \leq a < b \leq \theta$,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b \frac{1}{\theta}\, dx = \frac{b-a}{\theta}$$

- Height of the pdf is constant
- Probability is proportional to interval length

Let

$$X_1, \ldots, X_n \sim \text{Uniform}(0, \theta)$$

The parameter θ is **unknown**.

Natural Statistic: Sample Maximum

Since all observations lie in $[0, \theta]$,

$$\max(X_1, \ldots, X_n) \leq \theta$$

Define the statistic:

$$M_n = \max\{X_1, \ldots, X_n\}$$

- M_n contains information about θ
- Larger samples tend to produce larger M_n

Sampling Distribution of the Sample Maximum

For $0 \leq x \leq \theta$,

$$\mathbb{P}(M_n \leq x) = \mathbb{P}(X_1 \leq x, \ldots, X_n \leq x)$$

Using independence,

$$= \left(\frac{x}{\theta}\right)^n$$

Hence, the CDF of M_n is

$$F_{M_n}(x) = \begin{cases} \left(\frac{x}{\theta}\right)^n, & 0 \leq x \leq \theta \\ 1, & x \geq \theta \end{cases}$$

Using the Statistic to Estimate θ

From the distribution of M_n,

$$\mathbb{E}[M_n] = \frac{n}{n+1}\theta$$

This suggests the estimator:

$$\hat{\theta} = \frac{n+1}{n} M_n$$

- $\hat{\theta}$ is an unbiased estimator of θ
- Accuracy improves as sample size increases

Normal Distribution

A continuous random variable X is said to follow a **Normal**(θ, σ^2) distribution if its pdf is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\theta)^2}{2\sigma^2}\right), \quad -\infty < x < \infty$$

- θ is the mean (location parameter)
- Variance is fixed and equal to σ^2

$$\mathbb{E}[X] = \theta, \qquad \mathrm{Var}(X) = \sigma^2$$

Statistic for Normal(θ, σ^2) Data

Let

$$X_1, \ldots, X_n \sim \text{Normal}(\theta, \sigma^2)$$

The parameter θ is **unknown**.

- Goal: Estimate θ using the observed data
- Here we have assumed the population variance to be known and equal to σ^2 for simplicity.

Natural Statistic: Sample Mean

Define the sample mean:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

- $\bar{X}$ summarizes the central tendency of the data
- It is a natural candidate to estimate θ

Sampling Distribution of the Sample Mean

Since the Normal distribution is closed under averaging,

$$\bar{X} \sim \text{Normal}\left(\theta, \frac{\sigma^2}{n}\right)$$

Therefore,

$$\mathbb{E}[\bar{X}] = \theta, \qquad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

- $\bar{X}$ is an unbiased estimator of θ
- Variability decreases as n increases

Why Go Beyond One Random Variable?

So far, we studied **one random variable at a time**.

In practice, we often observe:

- Height and weight
- Input and output signals
- Arrival time and service time

These require studying **two or more random variables together**.

Bivariate Random Variables

A **bivariate random variable** is a pair

$$(X, Y)$$

defined on the same experiment.

- X and Y are observed together
- They may be related (dependent) or unrelated (independent)

Examples:

- $(X, Y) =$ (number of calls, waiting time)
- $(X, Y) =$ (temperature, power consumption)

Joint Distribution: Discrete Case

If X and Y are discrete, the joint pmf is

$$p_{X,Y}(x,y) = \mathbb{P}(X = x, Y = y)$$

Properties:

- $p_{X,Y}(x,y) \geq 0$
- $\sum_x \sum_y p_{X,Y}(x,y) = 1$

This fully describes the behavior of (X, Y).

Joint Distribution: Continuous Case

If X and Y are continuous, the joint pdf is

$$f_{X,Y}(x, y)$$

Properties:

- $f_{X,Y}(x, y) \geq 0$
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx\, dy = 1$

Probabilities are areas:

$$\mathbb{P}((X, Y) \in A) = \iint_A f_{X,Y}(x, y)\, dx\, dy$$

Marginal Distributions

Discrete case:

$$p_X(x) = \sum_y p_{X,Y}(x,y), \quad p_Y(y) = \sum_x p_{X,Y}(x,y)$$

Continuous case:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx$$

We "integrate out" the other variable.

Conditional Distributions

Discrete case:

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}, \quad p_Y(y) > 0$$

Continuous case:

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}, \quad f_Y(y) > 0$$

Conditional distributions are valid probability distributions.

Independence

X and Y are **independent** if:

$$p_{X,Y}(x,y) = p_X(x)p_Y(y)$$

(discrete)

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

(continuous)

Interpretation:

- Knowing Y gives no information about X

Example: Multinomial Distribution

The **Multinomial distribution** generalizes the Binomial case.

Suppose an experiment has k possible outcomes with probabilities

$$(p_1, p_2, \ldots, p_k), \quad \sum_{i=1}^{k} p_i = 1$$

Let

$$(X_1, \ldots, X_k) = \text{number of times each outcome occurs in } n \text{ trials}$$

Then,

$$(X_1, \ldots, X_k) \sim \text{Multinomial}(n; p_1, \ldots, p_k)$$

Joint pmf of the Multinomial Distribution

For $x_1 + \cdots + x_k = n$,

$$\mathbb{P}(X_1 = x_1, \ldots, X_k = x_k) = \frac{n!}{x_1! \cdots x_k!} \prod_{i=1}^{k} p_i^{x_i}$$

- Joint pmf of $(X_1, \ldots, X_k)$
- Marginals: $X_i \sim \text{Binomial}(n, p_i)$
- Components are **not independent**

Example: Uniform Distribution on the Unit Square

Let (X, Y) be uniformly distributed on the unit square:

$$0 \leq X \leq 1, \quad 0 \leq Y \leq 1$$

Joint pdf:

$$f_{X,Y}(x, y) = \begin{cases} 1, & 0 \leq x \leq 1,\ 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

This means every point in the unit square is equally likely.

Marginal and Conditional Distributions

Marginal distributions:

$$f_X(x) = \int_0^1 1\, dy = 1, \quad 0 \le x \le 1$$

$$f_Y(y) = \int_0^1 1\, dx = 1, \quad 0 \le y \le 1$$

Thus,

$$X, Y \sim \text{Uniform}(0, 1)$$

Conditional distribution:

$$f_{X|Y}(x|y) = 1, \quad 0 \le x \le 1$$

- X and Y are independent
- Joint = product of marginals

Example: Bivariate Normal Distribution

A pair (X, Y) follows a **bivariate normal distribution** if

$$(X, Y) \sim \mathcal{N}\left(\begin{pmatrix}\mu_X \\ \mu_Y\end{pmatrix}, \begin{pmatrix}\sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2\end{pmatrix}\right)$$

- $\rho \in [-1, 1]$ is the correlation coefficient
- Controls the dependence between X and Y

Properties of the Bivariate Normal

Joint density:

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)} \left\{ \left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 \right\} \right]$$

- Marginals:
$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$
- Conditional distributions:
$$X|Y = y \sim \mathcal{N}\left(\mu_X + \rho\frac{\sigma_X}{\sigma_Y}(y-\mu_Y), (1-\rho^2)\sigma_X^2\right)$$
- If $\rho = 0$, then X and Y are independent

Empirical Distribution Function

So far, we assumed a known distribution, but in practice the true distribution is often unknown. We need a **data-based** description of the distribution. Given observations

$$X_1, X_2, \ldots, X_n$$

We want to estimate the distribution function

$$F(x) = \mathbb{P}(X \leq x)$$

Empirical Distribution Function

So far, we assumed a known distribution, but in practice the true distribution is often unknown. We need a **data-based** description of the distribution. Given observations

$$X_1, X_2, \dots, X_n$$

We want to estimate the distribution function

$$F(x) = \mathbb{P}(X \leq x)$$

The idea:

- Count how many observations are $\leq x$
- Divide by total sample size

Empirical Distribution Function

The **empirical distribution function (EDF)** is defined as

$$F_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{X_i \leq x\}$$

- $\mathbf{1}\{\cdot\}$ is the indicator function
- $F_n(x)$ is the proportion of data $\leq x$

Properties of the Empirical Distribution Function

$F_n(x)$:

- Is a step function
- Increases only at observed data points
- Satisfies $0 \leq F_n(x) \leq 1$
- $\lim_{x \to -\infty} F_n(x) = 0$
- $\lim_{x \to +\infty} F_n(x) = 1$

It is a valid distribution function.

EDF is Itself a Random Function

- $F_n(x)$ depends on the sample
- Different samples $\rightarrow$ different EDFs

Hence, the EDF itself is a **random function**.

As sample size increases:

- $F_n(x)$ becomes more stable

EDF vs True Distribution

Let $F(x)$ be the true (unknown) distribution function.

- $F_n(x)$ estimates $F(x)$
- $F_n(x) \to F(x)$ as $n \to \infty$

This convergence holds:

- For each fixed x (by LLN)
- Uniformly over all x (Glivenko–Cantelli theorem)

Plug-In Principle: Basic Idea

Suppose a quantity of interest can be written as:

$$\theta = T(F)$$

where F is the (unknown) distribution function.

Plug-in principle:

$$\widehat{\theta} = T(F_n)$$

That is:

- Replace the true distribution F
- By the empirical distribution F_n
- Question: Why are these "good" estimates?

Example: Mean as a Plug-In Estimator

True mean:

$$\mu = \mathbb{E}[X] = \int x \, dF(x)$$

Plug in F_n:

$$\widehat{\mu} = \int x \, dF_n(x)$$

Since F_n places mass $1/n$ at each X_i:

$$\int x \, dF_n(x) = \frac{1}{n} \sum_{i=1}^{n} X_i$$

$$\boxed{\widehat{\mu} = \bar{X}}$$

Example: Second Moment as a Plug-In Estimator

True quantity:

$$\mathbb{E}[X^2] = \int x^2 \, dF(x)$$

Plug in F_n:

$$\int x^2 \, dF_n(x)$$

Since F_n places mass $1/n$ at each X_i:

$$\int x^2 \, dF_n(x) = \frac{1}{n} \sum_{i=1}^{n} X_i^2$$

$$\boxed{\widehat{\mathbb{E}[X^2]} = \frac{1}{n} \sum_{i=1}^{n} X_i^2}$$

Example: Variance as a Plug-In Estimator

True variance:

$$\sigma^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Plug in F_n:

$$\widehat{\sigma}^2 = \int x^2 \, dF_n(x) - \left(\int x \, dF_n(x) \right)^2$$

$$= \frac{1}{n} \sum_{i=1}^{n} X_i^2 - \bar{X}^2$$

Bias of the Plug-In Variance Estimator

Recall the plug-in estimator:

$$\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

Taking expectation:

$$\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \bar{X})^2\right] = (n-1)\sigma^2$$

Hence,

$$\mathbb{E}[\widehat{\sigma}^2] = \frac{n-1}{n}\sigma^2 \ < \ \sigma^2$$

- The plug-in estimator **underestimates** the true variance
- Bias arises because $\bar{X}$ is estimated from the same data
- Dividing by $n-1$ removes this bias

Example: Estimating a Probability

True quantity:

$$\mathbb{P}(X \leq x) = F(x)$$

Plug-in estimate:

$$\widehat{F}(x) = F_n(x)$$

This is exactly the empirical distribution function.

Thus: EDF is a plug-in estimator.

Example: Covariance as a Plug-In Estimator

True covariance:

$$\mathrm{Cov}(X, Y) = \int xy \, dF(x, y) - \left(\int x \, dF_X(x) \right) \left(\int y \, dF_Y(y) \right)$$

Plug in empirical distributions:

$$\int xy \, dF_n(x, y) = \frac{1}{n} \sum_{i=1}^{n} X_i Y_i$$

$$\int x \, dF_{n,X}(x) = \bar{X}, \qquad \int y \, dF_{n,Y}(y) = \bar{Y}$$

$$\boxed{\widehat{\mathrm{Cov}}(X, Y) = \frac{1}{n} \sum_{i=1}^{n} X_i Y_i - \bar{X} \bar{Y}}$$

Example: Correlation as a Plug-In Estimator

True correlation:

$$\rho = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}$$

Plug in empirical estimates:

$$\widehat{\rho} = \frac{\frac{1}{n}\sum_{i=1}^{n} X_i Y_i - \bar{X}\bar{Y}}{\sqrt{\left(\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}^2\right)\left(\frac{1}{n}\sum_{i=1}^{n} Y_i^2 - \bar{Y}^2\right)}}$$

Example: Quantile as a Plug-In Estimator

True p-quantile:

$$q_p = \inf\{x : F(x) \geq p\}$$

Plug in F_n:

$$\widehat{q}_p = \inf\{x : F_n(x) \geq p\}$$

If $X_{(1)} \leq \cdots \leq X_{(n)}$ are order statistics:

$$\boxed{\widehat{q}_p = X_{(\lceil np \rceil)}}$$

Question: For general F with density f, what is the sampling distribution of $X_{(k)}$?

Density of the kth Order Statistic

Idea: $X_{(k)} \in (x, x + dx)$ if:

- exactly $k - 1$ observations are $\leq x$,
- one observation falls in $(x, x + dx)$,
- the remaining $n - k$ observations are $> x$.

Probability:

$$\binom{n}{k-1,\ 1,\ n-k} [F(x)]^{k-1} f(x)\, dx\, [1 - F(x)]^{n-k}$$

Density of $X_{(k)}$:

$$\boxed{f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} [F(x)]^{k-1} [1 - F(x)]^{n-k} f(x)}$$

Example: Support Endpoint as a Plug-In Estimator

Right endpoint of the support:

$$\theta = \sup\{x : F(x) < 1\}$$

Plug in the empirical distribution function F_n:

$$\widehat{\theta} = \sup\{x : F_n(x) < 1\}$$

Since $F_n(x) = 1$ for all $x \geq \max(X_1, \ldots, X_n)$,

$$\boxed{\widehat{\theta} = \max(X_1, \ldots, X_n)}$$

As it can never reach theta it is always biased.
Question: Is there a constant we can use to make this unbiased?

$$\max(X_1, \ldots, X_n) \xrightarrow{a.s.} \theta$$

Example: Indicator Functional as a Plug-In Estimator

Parameter of interest:

$$\theta = \mathbf{1}\{\mathbb{E}[X] > c\}$$

Functional form:

$$\theta = \mathbf{1}\left\{\int x\, dF(x) > c\right\}$$

Plug in F_n:

$$\widehat{\theta} = \mathbf{1}\left\{\int x\, dF_n(x) > c\right\} = \mathbf{1}\{\bar{X} > c\}$$

Sampling variability:

$$\mathrm{Var}(\widehat{\theta}) = \mathbb{P}(\bar{X} > c)\big(1 - \mathbb{P}(\bar{X} > c)\big)$$

If $\mathbb{E}[X] = c$:

$$\lim_{n\to\infty} \mathrm{Var}(\widehat{\theta}) \neq 0$$

When Plug-In Estimators Can Fail (I)

Plug-in estimators do not always work well.

Case 1: Boundary / Extreme Parameters

- Example: right endpoint of support

$$\theta = \sup\{x : F(x) < 1\}$$

- Plug-in estimator is $\max X_i$
- Always underestimates θ

Case 2: Discontinuous Functionals

- Example:

$$\theta = \mathbb{I}\{\mathbb{E}[X] > 0\}$$

(We just want to know if the mean is positive or not)
- Plug-in estimator is unstable near the boundary

When Plug-In Estimators Can Fail (II)

Case 3: Bias in Finite Samples

- Example: variance estimation

$$\widehat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$$

- Plug-in estimator is biased

Case 4: Heavy-Tailed Distributions

- Mean may not exist (e.g., Cauchy)
- Sample mean becomes unstable

Summary: Limits of the Plug-In Principle

Plug-in estimators work well when:

- The parameter is smooth
- The distribution has finite moments
- Sample size is large

They may fail when:

- Parameters involve extremes or boundaries
- Functionals are discontinuous
- Data are heavy-tailed

Problem: Sampling Distribution of the EDF at a Fixed Point

Let

$$X_1, X_2, \ldots, X_n \sim F$$

be independent and identically distributed random variables with cumulative distribution function F.

The empirical distribution function (EDF) is defined as

$$F_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{X_i \leq x\}$$

Problem:

- Fix a point $x \in \mathbb{R}$.
- Find the sampling distribution of $F_n(x)$.
- Compute its mean and variance.

The Core Problem

We observe **only one sample**:

$$X_1, X_2, \ldots, X_n$$

But we want to understand:

- Bias of an estimator
- Variance of an estimator
- Sensitivity to individual observations

Problem:

- The true sampling distribution is unknown
- We cannot repeatedly sample from the population

Question: Can we reuse the data itself to approximate these quantities?

A Simple Idea: Leave-One-Out

Suppose our statistic is

$$\hat{\theta} = T(X_1, \ldots, X_n)$$

What happens if we remove one observation?

Define:

$$\hat{\theta}_{(-i)} = T(X_1, \ldots, X_{i-1}, X_{i+1}, \ldots, X_n)$$

This gives n slightly different estimates:

$$\hat{\theta}_{(-1)}, \hat{\theta}_{(-2)}, \ldots, \hat{\theta}_{(-n)}$$

These values reflect:

- Sampling variability
- Influence of each observation

The Jackknife Method

The **jackknife** is a resampling technique based on **leave-one-out samples**.

For a statistic $\hat{\theta} = T(X_1, \dots, X_n)$:

$$\hat{\theta}_{(-i)} = T(\text{sample with } X_i \text{ removed})$$

The **jackknife mean**:

$$\hat{\theta}_{\text{jack}} = \frac{1}{n} \sum_{i=1}^{n} \hat{\theta}_{(-i)}$$

This mimics repeated sampling using only the observed data.

Bias Depends on Sample Size

Let

$$\hat{\theta}_n = T(X_1, \ldots, X_n)$$

be a statistic estimating θ based on n observations.

For many smooth statistics, the bias admits an expansion:

$$\mathbb{E}[\hat{\theta}_n] = \theta + \frac{b}{n} + o\left(\frac{1}{n}\right)$$

- Bias typically decreases at rate $1/n$
- b is an unknown constant
- This holds for means, variances, smooth functionals

Question: Examples?

Effect of Removing One Observation

Define the leave-one-out estimator:

$$\hat{\theta}_{(-i)} = T(X_1, \ldots, X_{i-1}, X_{i+1}, \ldots, X_n)$$

which is based on $n - 1$ observations.

Its expectation satisfies:

$$\mathbb{E}[\hat{\theta}_{(-i)}] = \theta + \frac{b}{n-1} + o\left(\frac{1}{n}\right)$$

Let

$$\hat{\theta}_{\text{jack}} = \frac{1}{n}\sum_{i=1}^{n} \hat{\theta}_{(-i)}$$

Then,

$$\mathbb{E}[\hat{\theta}_{\text{jack}} - \hat{\theta}_n] = \frac{b}{n-1} - \frac{b}{n} = \frac{b}{n(n-1)}$$

Jackknife Bias Estimation

The jackknife estimate of bias is:

$$\widehat{\mathrm{Bias}}_{\mathsf{jack}} = (n-1)\left(\hat{\theta}_{\mathsf{jack}} - \hat{\theta}\right)$$

Bias-corrected estimator:

$$\hat{\theta}^{(\mathsf{bc})}_{\mathsf{jack}} = \hat{\theta} - \widehat{\mathrm{Bias}}_{\mathsf{jack}}$$

- Particularly useful when analytic bias is hard to compute
- Works well for smooth statistics

Example 1: Sample Variance (Biased Estimator)

Let $X_1, \ldots, X_n$ be i.i.d. with mean μ and variance σ^2.
Consider the estimator

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Although natural, this estimator is biased.

Theoretical Bias of $\hat{\sigma}^2$

Recall the identity:

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{i=1}^{n}(X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

Taking expectation,

$$\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \bar{X})^2\right] = n\sigma^2 - n\frac{\sigma^2}{n} = (n-1)\sigma^2.$$

Therefore,

$$\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2.$$

So the bias is $-\frac{\sigma^2}{n} = O\left(\frac{1}{n}\right)$.

Jackknife Leave-One-Out Variance

For the ith leave-one-out sample, define

$$\hat{\sigma}^2_{(-i)} = \frac{1}{n-1} \sum_{j \neq i} (X_j - \bar{X}_{(-i)})^2.$$

This has the same form as the original estimator, applied to $n - 1$ observations.

Jackknife Bias Estimator

The jackknife average is

$$\hat{\sigma}^2_{\text{jack}} = \frac{1}{n} \sum_{i=1}^{n} \hat{\sigma}^2_{(-i)}.$$

The jackknife estimate of bias is

$$\widehat{\text{Bias}}_{\text{jack}} = (n-1)\left(\hat{\sigma}^2_{\text{jack}} - \hat{\sigma}^2\right).$$

Jackknife Bias-Corrected Variance

The bias-corrected estimator is

$$\hat{\sigma}^{2(\mathrm{bc})}_{\mathsf{jack}} = \hat{\sigma}^2 - \widehat{\mathrm{Bias}}_{\mathsf{jack}}.$$

For the sample variance, this simplifies algebraically to

$$\hat{\sigma}^{2(\mathrm{bc})}_{\mathsf{jack}} = \frac{n}{n-1}\hat{\sigma}^2.$$

This coincides exactly with the classical unbiased estimator.

Example 2: Uniform$(0, \theta)$ Maximum Estimator

Let $X_1, \ldots, X_n \sim$ Uniform$(0, \theta)$.
A natural estimator of θ is:

$$\hat{\theta} = X_{(n)} = \max\{X_1, \ldots, X_n\}$$

We first compute its theoretical bias.

Theoretical Bias of $X_{(n)}$

The distribution of the maximum is:

$$F_{X_{(n)}}(x) = \left(\frac{x}{\theta}\right)^n, \quad 0 \le x \le \theta$$

Density:

$$f_{X_{(n)}}(x) = \frac{n}{\theta^n} x^{n-1}$$

Expectation:

$$\mathbb{E}[X_{(n)}] = \int_0^\theta x \frac{n}{\theta^n} x^{n-1} dx = \frac{n}{n+1}\theta$$

Hence,

$$\mathrm{Bias}(X_{(n)}) = \mathbb{E}[X_{(n)}] - \theta = -\frac{\theta}{n+1}$$

Jackknife Bias Estimation for $X_{(n)}$

Leave-one-out samples:

$$\hat{\theta}_{(-i)} = \max_{j \neq i} X_j$$

Jackknife average:

$$\hat{\theta}_{\text{jack}} = \frac{1}{n} \sum_{i=1}^{n} \hat{\theta}_{(-i)}$$

For the maximum:

- If $X_i \neq X_{(n)}$, then $\hat{\theta}_{(-i)} = X_{(n)}$
- If $X_i = X_{(n)}$, then $\hat{\theta}_{(-i)} = X_{(n-1)}$

Thus,

$$\hat{\theta}_{\text{jack}} = \frac{(n-1)X_{(n)} + X_{(n-1)}}{n}$$

Bias-Corrected Maximum Estimator

Jackknife bias estimate:

$$\widehat{\mathrm{Bias}}_{\mathsf{jack}} = (n-1)\left(\hat{\theta}_{\mathsf{jack}} - X_{(n)}\right) = \frac{n-1}{n}(X_{(n-1)} - X_{(n)})$$

Bias-corrected estimator :

$$\hat{\theta}^{(\mathsf{bc})}_{\mathsf{jack}} = X_{(n)} - \widehat{\mathrm{Bias}}_{\mathsf{jack}} = \frac{2n-1}{n}X_{(n)} - \frac{n-1}{n}X_{(n-1)}$$

This estimator significantly reduces finite-sample bias.

Expectation of the Bias-Corrected Estimator (Uniform Case)

Recall the jackknife bias-corrected estimator:

$$\hat{\theta}_{\text{jack}}^{(\text{bc})} = \frac{2n-1}{n} X_{(n)} - \frac{n-1}{n} X_{(n-1)}$$

We use the known expectations:

$$\mathbb{E}[X_{(n)}] = \frac{n}{n+1}\theta, \qquad \mathbb{E}[X_{(n-1)}] = \frac{n-1}{n+1}\theta$$

Therefore,

$$\mathbb{E}\left[\hat{\theta}_{\text{jack}}^{(\text{bc})}\right] = \frac{2n-1}{n} \cdot \frac{n}{n+1}\theta - \frac{n-1}{n} \cdot \frac{n-1}{n+1}\theta$$

$$= \theta\left(1 - \frac{1}{n(n+1)}\right) = \theta - O\left(\frac{1}{n^2}\right)$$

Example 3: Bernoulli(p) Functional Estimation

Let $X_1, \ldots, X_n \sim \mathrm{Bernoulli}(p)$.
Recall:

$$\mathbb{E}[X] = p, \qquad \mathrm{Var}(X) = p(1-p).$$

The natural estimator of p is:

$$\hat{p} = \bar{X}$$

We consider estimation of the functional:

$$\theta = p^2$$

Theoretical Bias of $\hat{\theta} = \hat{p}^2$

We have:

$$\mathbb{E}[\hat{p}] = p, \qquad \mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

Hence,

$$\mathbb{E}[\hat{p}^2] = \mathrm{Var}(\hat{p}) + \{\mathbb{E}[\hat{p}]\}^2 = p^2 + \frac{p(1-p)}{n}.$$

Therefore,

$$\mathrm{Bias}(\hat{p}^2) = \frac{p(1-p)}{n} = O\left(\frac{1}{n}\right).$$

Jackknife Leave-One-Out Estimators

Leave-one-out mean:

$$\hat{p}_{(-i)} = \frac{1}{n-1}\sum_{j\neq i} X_j = \frac{n\hat{p} - X_i}{n-1}.$$

Leave-one-out estimator:

$$\hat{\theta}_{(-i)} = \hat{p}^2_{(-i)}.$$

Jackknife average:

$$\hat{\theta}_{\text{jack}} = \frac{1}{n}\sum_{i=1}^{n} \hat{p}^2_{(-i)}.$$

Jackknife Bias Calculation

Using algebra,

$$\frac{1}{n}\sum_{i=1}^{n}\hat{p}_{(-i)}^{2} = \hat{p}^{2} + \frac{1}{(n-1)^{2}}\hat{p}(1-\hat{p}).$$

Therefore, the jackknife bias estimate is:

$$\widehat{\mathrm{Bias}}_{\text{jack}} = (n-1)\left(\hat{\theta}_{\text{jack}} - \hat{p}^{2}\right) = \frac{1}{n-1}\hat{p}(1-\hat{p}).$$

Bias-Corrected Estimator

The jackknife bias-corrected estimator is:

$$\hat{\theta}^{(\mathrm{bc})}_{\mathsf{jack}} = \hat{p}^2 - \widehat{\mathrm{Bias}}_{\mathsf{jack}} = \hat{p}^2 - \frac{1}{n-1}\hat{p}(1-\hat{p}).$$

Taking expectation,

$$\mathbb{E}\left[\hat{\theta}^{(\mathrm{bc})}_{\mathsf{jack}}\right] = p^2.$$

The $O(1/n)$ bias is removed exactly.

Check: Jackknife bias-corrected estimate removes bias for Poisson(λ) when trying to estimate λ^2 using $\bar{X}^2$

From Bias to Variance Estimation

So far, we used the jackknife to:

- Estimate bias of an estimator
- Construct bias-corrected estimators

Next question:

How do we estimate the **variance** of a statistic

$$\hat{\theta} = T(X_1, \dots, X_n)$$

when the true sampling distribution is unknown?

Why Variance Estimation is Hard

Recall:

$$\mathrm{Var}(\hat{\theta}) = \mathbb{E}\left[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2\right]$$

In practice:

- We observe only one dataset
- The distribution of $\hat{\theta}$ is unknown
- Analytic variance formulas may not exist

Goal: Estimate $\mathrm{Var}(\hat{\theta})$ using only the observed data.

Leave-One-Out Estimates

Define the leave-one-out estimators:

$$\hat{\theta}_{(-i)} = T(X_1, \ldots, X_{i-1}, X_{i+1}, \ldots, X_n), \quad i = 1, \ldots, n$$

These are:

- Slight perturbations of $\hat{\theta}$
- Based on samples of size $n-1$
- Proxies for repeated sampling

Definition: Jackknife Variance Estimator

The **jackknife variance estimator** of $\hat{\theta}$ is defined as:

$$\boxed{\widehat{\mathrm{Var}}_{\text{jack}}(\hat{\theta}) = \frac{n-1}{n}\sum_{i=1}^{n}(\hat{\theta}_{(-i)} - \hat{\theta}_{\text{jack}})^2}$$

where

$$\hat{\theta}_{\text{jack}} = \frac{1}{n}\sum_{i=1}^{n}\hat{\theta}_{(-i)}.$$

Where Does This Formula Come From?

To understand the structure of the jackknife variance, we introduce an alternative representation.
Define the **jackknife pseudo-values**:

$$\boxed{\theta_i^* = n\hat{\theta} - (n-1)\hat{\theta}_{(-i)}, \quad i = 1, \ldots, n}$$

Interpretation:

- Each θ_i^* behaves like a bias-corrected estimate of θ
- Removes the leading $O(1/n)$ bias
- Acts like a "corrected data point"

Pseudo-Values as Corrected Observations

Key properties of pseudo-values:

- $\theta_1^*, \ldots, \theta_n^*$ play the role of a sample
- Each observation contributes one pseudo-value
- Bias from estimating θ is corrected to first order

The jackknife bias corrected estimator of θ can be written as:

$$\boxed{\hat{\theta}_{\text{jack}}^{bc} = \frac{1}{n}\sum_{i=1}^{n}\theta_i^*} = n\hat{\theta} - (n-1)\hat{\theta}_{jack}$$

This mirrors the sample mean of data.

Variance from Pseudo-Values

Since $\theta_1^*, \ldots, \theta_n^*$ behave like corrected observations, we estimate their variability using the sample variance:

$$s_*^2 = \frac{1}{n-1} \sum_{i=1}^{n} (\theta_i^* - \hat{\theta}_{\text{jack}}^{bc})^2$$

But $\hat{\theta}_{\text{jack}}^{bc}$ is the mean of n observations.

Therefore:

$$\boxed{\widehat{\mathrm{Var}}_{\text{jack}}(\hat{\theta}) = \frac{s_*^2}{n}}$$

Final Jackknife Variance Formula

Substituting the definition of θ_i^*,

$$\begin{aligned}
\widehat{\mathrm{Var}}_{\text{jack}}(\hat{\theta}) &= \frac{1}{n(n-1)} \sum_{i=1}^{n} (\theta_i^* - \hat{\theta}_{\text{jack}}^{bc})^2 \\
&= \frac{1}{n(n-1)} \sum_{i=1}^{n} (n-1)^2 (\hat{\theta}_{(-i)} - \hat{\theta}_{\text{jack}})^2 \\
&= \frac{n-1}{n} \sum_{i=1}^{n} (\hat{\theta}_{(-i)} - \hat{\theta}_{\text{jack}})^2
\end{aligned}$$

Two equivalent views:

- Variance of leave-one-out estimates
- Variance of pseudo-values divided by n

Example 1: Sample Mean

Let

$$\hat{\theta} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

Leave-one-out mean:

$$\bar{X}_{(-i)} = \frac{n\bar{X} - X_i}{n-1}$$

Jackknife mean:

$$\bar{X}_{\text{jack}} = \bar{X}$$

Plugging into the variance formula:

$$\widehat{\mathrm{Var}}_{\text{jack}}(\bar{X}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

This equals the classical variance estimator of $\bar{X}$.

Example 2: Sample Variance

Let

$$X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2).$$

Consider the statistic:

$$\hat{\theta} = \hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

Recall:

- $\hat{\sigma}^2$ is unbiased for σ^2
- $\hat{\sigma}^2$ is a nonlinear statistic

Exact Distributional Facts (Normal Case)

Under normality:

$$\frac{(n-1)\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-1}.$$

Hence,

$$\mathbb{E}[\hat{\sigma}^2] = \sigma^2, \qquad \mathrm{Var}(\hat{\sigma}^2) = \frac{2}{n-1}\sigma^4.$$

This is the **true sampling variance**.

Leave-One-Out Sample Variances

Define the leave-one-out estimators:

$$\hat{\sigma}^2_{(-i)} = \frac{1}{n-2} \sum_{j \neq i} (X_j - \bar{X}_{(-i)})^2.$$

The jackknife mean is:

$$\hat{\sigma}^2_{\text{jack}} = \frac{1}{n} \sum_{i=1}^{n} \hat{\sigma}^2_{(-i)}.$$

The jackknife variance estimator is:

$$\widehat{\mathrm{Var}}_{\text{jack}}(\hat{\sigma}^2) = \frac{n-1}{n} \sum_{i=1}^{n} (\hat{\sigma}^2_{(-i)} - \hat{\sigma}^2_{\text{jack}})^2 = \frac{n^2}{(n-1)(n-2)^2} (\hat{\mu}_4 - \hat{\mu}_2^2)$$

Expectation of the Jackknife Variance

Where :

$$\hat{\mu}_k = \frac{1}{n}\sum_i (X_i - \bar{X})^2.$$

Check that:

$$\mathbb{E}\left[\widehat{\mathrm{Var}}_{\mathsf{jack}}(\hat{\sigma}^2)\right] = \frac{2\sigma^4}{n-2}$$

$$\widehat{\mathrm{Var}}_{\mathsf{jack}}(\hat{\sigma}^2) \xrightarrow{P} \mathrm{Var}(\hat{\sigma}^2).$$

Thus:

$$\boxed{\widehat{\mathrm{Var}}_{\mathsf{jack}}(\hat{\sigma}^2) \text{ is biased but consistent}}$$

Example 3: Jackknife Variance for the Sample Maximum

Let

$$X_1, \ldots, X_n \sim \mathrm{Uniform}(0, 1),$$

and consider the estimator

$$\hat{\theta} = X_{(n)} = \max\{X_1, \ldots, X_n\}.$$

We study the jackknife variance estimator for $\hat{\theta}$.

True Variance of the Maximum

The distribution of $X_{(n)}$ is:

$$F_{X_{(n)}}(x) = x^n, \quad 0 \leq x \leq 1.$$

Moments:

$$\mathbb{E}[X_{(n)}] = \frac{n}{n+1}, \qquad \mathrm{Var}(X_{(n)}) = \frac{n}{(n+1)^2(n+2)}.$$

Hence,

$$\boxed{n^2 \mathrm{Var}(X_{(n)}) \longrightarrow 1}$$

Leave-One-Out Maxima

Define the leave-one-out estimator:

$$\hat{\theta}_{(-i)} = \max_{j \neq i} X_j.$$

There are only two possibilities:

- If $X_i \neq X_{(n)}$, then

$$\hat{\theta}_{(-i)} = X_{(n)}$$

- If $X_i = X_{(n)}$, then

$$\hat{\theta}_{(-i)} = X_{(n-1)}$$

Thus:

$$\hat{\theta}_{(-1)}, \ldots, \hat{\theta}_{(-n)} = \underbrace{X_{(n)}, \ldots, X_{(n)}}_{n-1 \text{ times}},\ X_{(n-1)}.$$

Jackknife Mean

The jackknife mean is:

$$\hat{\theta}_{\text{jack}} = \frac{1}{n}\Big[(n-1)X_{(n)} + X_{(n-1)}\Big].$$

Equivalently,

$$\hat{\theta}_{\text{jack}} = X_{(n)} - \frac{1}{n}\big(X_{(n)} - X_{(n-1)}\big).$$

Jackknife Variance: Exact Formula

The jackknife variance estimator is:

$$\widehat{\mathrm{Var}}_{\text{jack}}(X_{(n)}) = \frac{n-1}{n} \sum_{i=1}^{n} (\hat{\theta}_{(-i)} - \hat{\theta}_{\text{jack}})^2.$$

Compute deviations:

- For $n-1$ terms:

$$X_{(n)} - \hat{\theta}_{\text{jack}} = \frac{1}{n}\left(X_{(n)} - X_{(n-1)}\right)$$

- For 1 term:

$$X_{(n-1)} - \hat{\theta}_{\text{jack}} = -\frac{n-1}{n}\left(X_{(n)} - X_{(n-1)}\right)$$

Simplifying the Jackknife Variance

Summing the squared deviations:

$$\sum_{i=1}^{n}(\cdot)^2 = (n-1)\left(\frac{\Delta_n}{n}\right)^2 + \left(\frac{(n-1)\Delta_n}{n}\right)^2,$$

where

$$\Delta_n = X_{(n)} - X_{(n-1)}.$$

Therefore,

$$\boxed{\widehat{\mathrm{Var}}_{\text{jack}}(X_{(n)}) = \frac{(n-1)^2}{n^2}\,\Delta_n^2}$$

Key Quantity: The Extreme Spacing

To understand the jackknife variance, we must study

$$\Delta_n = X_{(n)} - X_{(n-1)}.$$

Claim:

$$\boxed{n\Delta_n \xrightarrow{d} \mathrm{Exp}(1)}$$

We now prove this result.

Rewriting the Probability

Fix $t \geq 0$.

$$\mathbb{P}(n\Delta_n > t) = \mathbb{P}\left(X_{(n)} - X_{(n-1)} > \frac{t}{n}\right).$$

This event means:

No observation lies in the interval

$$\left(X_{(n)} - \frac{t}{n}, X_{(n)}\right).$$

Conditioning on the Maximum

Condition on the value of the maximum:

$$X_{(n)} = x.$$

Given this:

- The remaining $n - 1$ points are independent
- They are uniformly distributed on $(0, x)$

Probability of No Points Near the Maximum

Given $X_{(n)} = x$:

$$\mathbb{P}\left(X_{(n-1)} \le x - \frac{t}{n} \,\middle|\, X_{(n)} = x\right) = \left(1 - \frac{t}{nx}\right)^{n-1}.$$

As $n \to \infty$:

$$\left(1 - \frac{t}{nx}\right)^{n-1} \longrightarrow e^{-t/x}.$$

Removing the Conditioning

For Uniform$(0, 1)$:

$$X_{(n)} \xrightarrow{P} 1.$$

Hence,

$$e^{-t/X_{(n)}} \xrightarrow{P} e^{-t}.$$

Therefore,

$$\mathbb{P}(n\Delta_n > t) \longrightarrow e^{-t}.$$

Exponential Limit and Consequence

We have shown:

$$n\big(X_{(n)} - X_{(n-1)}\big) \xrightarrow{d} \mathrm{Exp}(1).$$

where $\xrightarrow{d}$ means "has limiting distribution". Thus,

$$n^2\widehat{\mathrm{Var}}_{\mathsf{jack}}(X_{(n)}) \xrightarrow{d} (\mathrm{Exp}(1))^2,$$

a non-degenerate random variable.

$\boxed{\text{Jackknife variance does not converge to 0}}$

Conclusion

- True variance of $X_{(n)}$ is $O(1/n^2)$
- Jackknife variance converges to a random limit
- Hence jackknife variance is **inconsistent**

$\boxed{\text{Jackknife variance fails for the sample maximum}}$

From Jackknife to Bootstrap

Recall the jackknife:

- Uses deterministic leave-one-out samples
- Produces n pseudo-samples of size $n-1$
- Approximates bias via first-order expansions

Limitation:

- Jackknife approximates only **local** behavior of T
- Fails for non-smooth or boundary functionals

Bootstrap idea:

Approximate the entire sampling distribution of T_n

Statistical Setting

Let

$$X_1, \ldots, X_n \sim F$$

be i.i.d., where F is an unknown distribution on $\mathbb{R}$.

Let

$$T_n = T(X_1, \ldots, X_n)$$

be a statistic estimating

$$\theta = T(F).$$

Goal:

$$\text{Approximate } \mathcal{L}_F(T_n) \quad \text{using only } X_1, \ldots, X_n.$$

Here $\mathcal{L}_F(T_n)$ means distribution of T_n when $X_i \sim F$

Empirical Distribution as a Random Distribution

The empirical distribution function is

$$F_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{X_i \le x\}.$$

F_n is a **random probability distribution** satisfying:

$$F_n \xrightarrow{a.s.} F \quad \text{(Glivenko–Cantelli)}.$$

Bootstrap replaces F by F_n at the distributional level.

Bootstrap Principle (Formal Statement)

Bootstrap approximation:

$$\boxed{\mathcal{L}_F(T(F)) \;\approx\; \mathcal{L}_{F_n}(T(F_n))}$$

Interpreted as:

- Real world law: $\mathcal{L}_F(T_n)$
- Bootstrap world law: $\mathcal{L}_{F_n}(T_n^*)$

Approximation is conditional on the observed data.

Bootstrap World: Conditional Probability

Define bootstrap samples:

$$X_1^*, \ldots, X_n^* \mid X_1, \ldots, X_n \overset{i.i.d.}{\sim} F_n.$$

Equivalently,

$$X_j^* = X_{I_j}, \qquad I_j \sim \mathrm{Unif}\{1, \ldots, n\},$$

independently.

Bootstrap statistic:

$$T_n^* = T(X_1^*, \ldots, X_n^*).$$

All probabilities below are **conditional on the data**.

Bootstrap Law and Notation

Define:

$$\mathbb{P}^*(\cdot) = \mathbb{P}(\cdot \mid X_1, \ldots, X_n), \quad \mathbb{E}^*(\cdot), \quad \mathrm{Var}^*(\cdot).$$

Bootstrap law:

$$\mathcal{L}^*(T_n^*) = \mathcal{L}(T_n^* \mid X_1, \ldots, X_n).$$

This is a random distribution depending on the sample.

Bootstrap Algorithm (Mathematical Form)

Given data $X_1, \ldots, X_n$:

1. Compute $T_n = T(X_1, \ldots, X_n)$
2. Generate i.i.d. copies

$$T_n^{*(1)}, \ldots, T_n^{*(B)} \sim \mathcal{L}^*(T_n^*)$$

Approximate:

$$\mathcal{L}_F(T_n) \approx \widehat{\mathcal{L}}^*(T_n^*)$$

(empirical distribution of the bootstrap replicates).

Bootstrap Bias Estimation

True bias:

$$\mathrm{Bias}(T_n) = \mathbb{E}_F[T_n] - \theta.$$

Bootstrap bias estimator:

$$\widehat{\mathrm{Bias}}_{\mathsf{boot}} = \mathbb{E}^*[T_n^*] - T_n.$$

Monte Carlo approximation:

$$\mathbb{E}^*[T_n^*] \approx \frac{1}{B} \sum_{b=1}^{B} T_n^{*(b)}.$$

Bootstrap Variance Estimation

True variance:

$$\mathrm{Var}_F(T_n).$$

Bootstrap variance:

$$\widehat{\mathrm{Var}}_{\mathrm{boot}}(T_n) = \mathrm{Var}^*(T_n^*).$$

Estimated by:

$$\frac{1}{B-1}\sum_{b=1}^{B}\left(T_n^{*(b)} - \bar{T}_n^*\right)^2, \quad \bar{T}_n^* = \frac{1}{B}\sum_{b=1}^{B} T_n^{*(b)}.$$

Example: Sample Mean (Bootstrap is Exact)

Let

$$T_n = \bar{X}.$$

Conditionally on the data:

$$\mathbb{E}^*[\bar{X}^*] = \bar{X}, \qquad \mathrm{Var}^*(\bar{X}^*) = \frac{s_n^2}{n},$$

where

$$s_n^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Bootstrap reproduces the exact first-order sampling behavior.

Example: Sample Variance (Bootstrap Bias)

Let

$$T_n = S_n^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Bootstrap version:

$$S_n^{*2} = \frac{1}{n}\sum_{j=1}^{n}(X_j^* - \bar{X}^*)^2 = \overline{X^{*2}} - (\bar{X}^*)^2.$$

Conditionally on the data:

$$\mathbb{E}^*(X^{*2}) = \frac{1}{n}\sum_{i=1}^{n} X_i^2, \qquad \mathbb{E}^*(\bar{X}^*) = \bar{X}, \qquad \mathrm{Var}^*(\bar{X}^*) = \frac{s_n^2}{n}.$$

Hence,

$$\mathbb{E}^*(S_n^{*2}) = \mathbb{E}^*(X^{*2}) - \mathbb{E}^*(\bar{X}^{*2}) = \left(1 - \frac{1}{n}\right) s_n^2.$$

Example: Sample Maximum (Bootstrap Failure)

Let

$$T_n = M_n = \max(X_1, \ldots, X_n), \qquad X_i \sim \mathrm{Uniform}(0, \theta).$$

True sampling behavior:

$$n(\theta - M_n) \;\Rightarrow\; \mathrm{Exp}(\theta).$$

Bootstrap world:

$$\mathrm{supp}(F_n) = [0, M_n]$$

$$P(M_n > M_n^*) = \left(\frac{n-1}{n}\right)^n \to \frac{1}{e} \qquad \text{as } n \to \infty$$

$$P(n(M_n - M_n^*) = 0) \to 1 - \frac{1}{e}$$

Which does not agree with true behavior. Hence Bootstrap fails to reproduce the correct limiting distribution.

Why Bootstrap Fails for Extremes

The parameter

$$\theta = \sup\{x : F(x) < 1\}$$

is:

- A boundary functional
- Not smooth

Bootstrap fails when:

- Statistic depends on extreme order statistics
- Limiting law is non-Gaussian

Question: What do we mean by *non-smooth* parameter?

Influence Function

Let $T(F)$ be a parameter defined as a functional of a distribution F.

For any point x, define the contaminated distribution:

$$F_\varepsilon = (1 - \varepsilon)F + \varepsilon\delta_x,$$

where δ_x puts unit mass at x.

The **influence function** of T at F is:

$$\boxed{\mathrm{IF}(x; T, F) = \left.\frac{d}{d\varepsilon} T(F_\varepsilon)\right|_{\varepsilon=0}}$$

Interpretation of the Influence Function

$$\mathrm{IF}(x; T, F) = \text{first-order effect of adding a point at } x$$

Interpretation:

- Measures sensitivity of $T(F)$ to an outlier at x
- Large values $\Rightarrow$ estimator is sensitive
- Bounded IF $\Rightarrow$ robust estimator

Influence functions are properties of the **functional** T, not the data.

Example: Influence Function of the Mean

Parameter:

$$T(F) = \mu = \int x \, dF(x)$$

For $F_\varepsilon = (1-\varepsilon)F + \varepsilon\delta_x$:

$$T(F_\varepsilon) = (1-\varepsilon)\mu + \varepsilon x$$

Hence:

$$\mathrm{IF}(x; T, F) = \frac{d}{d\varepsilon} T(F_\varepsilon)\bigg|_{\varepsilon=0} = x - \mu.$$

Example: Influence Function of the Variance

Parameter:

$$T(F) = \sigma^2 = \int (x-\mu)^2 \, dF(x)$$

The influence function is:

$$\mathrm{IF}(x; T, F) = (x-\mu)^2 - \sigma^2.$$

Thus:

- Mean: linear IF
- Variance: quadratic IF

From Influence Function to Linearization

If T has an influence function $\mathrm{IF}(x; T, F)$, then:

$$T(F_n) - T(F) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{IF}(X_i; T, F) + o_p(n^{-1/2}).$$

This is the **linearization** used in:

- Asymptotic normality
- Jackknife
- Bootstrap consistency

From Influence Function to Linearization(**)

Suppose G is a distribution which is close to F. Then, in some sense, we should be able to write:

$$T(G) - T(F) \approx T^{'}(F)(G - F)$$

Now consider any set A. Observe:

$$\int_x \delta_x(A)dG = G(A) \text{ and } \int_x F(A)dG = F(A)$$

$$\implies \int_x (\delta_x(A) - F(A))dG = G(A) - F(A) \forall A$$

$$\implies \int_x (\delta_x - F)dG = G - F$$

$$\implies T(G) - T(F) \approx \int T^{'}(F)(\delta_x - F)dG$$

From Influence Function to Linearization(**)

Now from the definition of influence function, we know $F_\epsilon = (1-\epsilon)F + \epsilon\delta_x = F + \epsilon(\delta_x - F)$, Hence:

$$\frac{T(F_\epsilon) - T(F)}{\epsilon} \approx T^{'}(F)(\delta_x - F)$$

Substituting this we get :

$$T(G) - T(F) \approx \int IF(x, T, F)dG$$

if we take $G = F_n$ we reach the linearization condition.
Note : Taking $G = F$ shows that the expected value of the influence function is zero.

Why Influence Functions Matter

Let:

$$\psi(x) = \mathrm{IF}(x; T, F).$$

Then:

$$\sqrt{n}(T_n - \theta) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi(X_i) + o_p(1).$$

Consequences:

- CLT applies if $\mathbb{E}[\psi(X)^2] < \infty$
- Bootstrap reproduces this distribution

When Does Linearization Hold?

The expansion

$$T(F_n) - T(F) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{IF}(X_i; T, F) + o_p(n^{-1/2})$$

is called a **linearization**.

This linearization:

- Holds for many *smooth* statistical functionals
- Is typically derived using Taylor expansion arguments
- Fails for non-smooth or boundary parameters

Example 1: Sample Mean

Parameter:

$$T(F) = \mu = \int x \, dF(x)$$

Influence function:

$$\mathrm{IF}(x; T, F) = x - \mu$$

Linearization:

$$\bar{X} - \mu = \frac{1}{n} \sum_{i=1}^{n} (X_i - \mu)$$

Here the linearization is **exact** (no remainder term).

Example 2: Sample Variance

Parameter:

$$T(F) = \sigma^2$$

Influence function:

$$\mathrm{IF}(x; T, F) = (x - \mu)^2 - \sigma^2$$

Linearization:

$$\hat{\sigma}^2 - \sigma^2 = \frac{1}{n} \sum_{i=1}^{n} \left[(X_i - \mu)^2 - \sigma^2 \right] + o_p(n^{-1/2})$$

The remainder arises due to estimation of μ.

Example 3: Smooth Function of the Mean

Let

$$T(F) = g(\mu), \qquad \mu = \int x \, dF(x),$$

where g is continuously differentiable.

By Taylor expansion:

$$g(\bar{X}) - g(\mu) = g'(\mu)(\bar{X} - \mu) + o_p(n^{-1/2})$$

Thus:

$$\mathrm{IF}(x; T, F) = g'(\mu)(x - \mu)$$

Smooth transformations preserve linearization.

Typical Situations Where Linearization Holds

Linearization usually holds for:

- Means and moments
- Variances and covariances
- Smooth functions of sample averages

In all these cases:

$$T_n - \theta = O_p(n^{-1/2})$$

and CLT-type behavior emerges.

When Linearization Fails

Linearization may fail when:

- The functional is non-smooth
- The parameter lies on the boundary
- Small changes in data cause large jumps in T_n

Examples:

- Sample maximum
- Support endpoint estimation

Outline: Proof of Bootstrap Consistency

We outline why the bootstrap consistently estimates the sampling distribution of T_n.

Goal:

$$\mathcal{L}\big(\sqrt{n}(T_n^* - T_n) \mid X_1, \ldots, X_n\big) \implies \mathcal{L}\big(\sqrt{n}(T_n - \theta)\big)$$

in probability.

Key Assumption: Linearization

Assume the statistic admits a first-order expansion:

$$T_n - \theta = \frac{1}{n}\sum_{i=1}^{n} \psi(X_i) + r_n,$$

where:

- ψ is the **influence function**
- $\mathbb{E}[\psi(X)] = 0$
- $\mathbb{E}[\psi(X)^2] < \infty$
- $r_n = o_p(n^{-1/2})$

This is analogous to a Taylor expansion.

Step 1: Limiting Distribution in the Real World

Multiply the expansion by $\sqrt{n}$:

$$\sqrt{n}(T_n - \theta) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi(X_i) + o_p(1).$$

By the Central Limit Theorem:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi(X_i) \xrightarrow{d} \mathcal{N}(0, \sigma^2), \quad \sigma^2 = \mathrm{Var}(\psi(X)).$$

Thus,

$$\sqrt{n}(T_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

Step 2: Bootstrap Expansion

Condition on the observed data $X_1, \dots, X_n$.

Bootstrap sample:

$$X_1^*, \dots, X_n^* \sim F_n \quad \text{i.i.d.}$$

Using the same linearization:

$$T_n^* - T_n = \frac{1}{n}\sum_{i=1}^{n}\left(\psi(X_i^*) - \bar{\psi}_n\right) + r_n^*,$$

where:

$$\bar{\psi}_n = \frac{1}{n}\sum_{i=1}^{n}\psi(X_i), \quad r_n^* = o_p^*(n^{-1/2}).$$

Step 3: CLT (Bootstrap World)

Conditionally on the data:

- $\psi(X_1^*), \ldots, \psi(X_n^*)$ are i.i.d. from the empirical distribution
- Mean $= \bar{\psi}_n$
- Variance $\frac{1}{n} \sum_{i-1=1}^{n} (\psi(X_i) - \bar{\psi}_n)^2 \to \sigma^2$

By the CLT:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \left(\psi(X_i^*) - \bar{\psi}_n\right) \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

Step 4: Matching the Limits

We have shown:

$$\sqrt{n}(T_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2),$$

and

$$\sqrt{n}(T_n^* - T_n) \,\Big|\, X_1, \ldots, X_n \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

Therefore, the bootstrap reproduces the correct limit law.

Conclusion: Bootstrap Consistency

$$\boxed{\mathcal{L}\left(\sqrt{n}(T_n^* - T_n) \mid X_1, \ldots, X_n\right) \implies \mathcal{L}\left(\sqrt{n}(T_n - \theta)\right)}$$

The bootstrap consistently estimates the sampling distribution of T_n.

When This Argument Fails

The argument relies on the linearization step.

Bootstrap may fail when:

- No first-order expansion exists
- Remainder r_n is not $o_p(n^{-1/2})$
- Statistic depends on extremes or boundaries

Examples:

- Sample maximum
- Support endpoint estimation

Influence Function: Sample Median

Let

$$T(F) = F^{-1}(1/2), \qquad m = F^{-1}(1/2).$$

The influence function at point x is defined as

$$\mathrm{IF}(x; T, F) = \lim_{\varepsilon \downarrow 0} \frac{T\big((1-\varepsilon)F + \varepsilon\delta_x\big) - T(F)}{\varepsilon}.$$

Assume F has density f with $f(m) > 0$.

Exact Calculation: Sample Median

Let

$$F_\varepsilon = (1-\varepsilon)F + \varepsilon\delta_x, \qquad m_\varepsilon = T(F_\varepsilon).$$

By definition,

$$(1-\varepsilon)F(m_\varepsilon) + \varepsilon\mathbf{1}\{x \le m_\varepsilon\} = \tfrac{1}{2}.$$

Using $F(m) = \frac{1}{2}$ and Taylor expansion,

$$F(m_\varepsilon) = \tfrac{1}{2} + f(m)(m_\varepsilon - m) + o(m_\varepsilon - m).$$

Exact Calculation: Sample Median (continued)

Substituting and collecting $O(\varepsilon)$ terms,

$$f(m)(m_\varepsilon - m) = -\varepsilon\left(\mathbf{1}\{x \le m\} - \tfrac{1}{2}\right) + o(\varepsilon).$$

Hence,

$$m_\varepsilon - m = -\varepsilon \frac{\mathbf{1}\{x \le m\} - 1/2}{f(m)} + o(\varepsilon).$$

Influence Function: Sample Median (Conclusion)

$$\boxed{\mathrm{IF}(x; T, F) = \frac{\mathbf{1}\{x \leq m\} - 1/2}{f(m)}}$$

- IF is bounded
- Linearization exists only if $f(m) > 0$
- Bootstrap validity depends on local smoothness

Failure of Linearization: Sample Maximum

Let

$$T(F) = \sup\{x : F(x) < 1\} = \theta.$$

For any signed perturbation h,

$$\frac{T(F + th) - T(F)}{t}$$

does not converge as $t \downarrow 0$.

Exact Calculation: Sample Maximum

Let

$$F_\varepsilon = (1-\varepsilon)F + \varepsilon\delta_x.$$

If $x \leq \theta$,

$$T(F_\varepsilon) = \theta.$$

If $x > \theta$,

$$T(F_\varepsilon) = x.$$

Exact Calculation: Sample Maximum (continued)

Therefore,

$$\frac{T(F_\varepsilon) - T(F)}{\varepsilon} = \begin{cases} 0, & x \leq \theta, \\ \dfrac{x-\theta}{\varepsilon}, & x > \theta. \end{cases}$$

For $x > \theta$,

$$\lim_{\varepsilon \downarrow 0} \frac{x-\theta}{\varepsilon} = \infty.$$

Failure of Linearization: Sample Maximum (Conclusion)

Small changes in F near the boundary produce $O(1)$ changes in $T(F)$.

- No linear first-order expansion exists
- r_n is not $o_p(n^{-1/2})$
- Bootstrap fails even asymptotically

Extreme-value statistics violate the bootstrap linearization principle.

Why Parametric Bootstrap?

Nonparametric bootstrap (resampling from F_n) fails when:

- The parameter is a boundary functional (e.g., support endpoint)
- The statistic depends on extreme order statistics
- The functional is non-smooth (no influence function)
- The limit law is non-Gaussian

Idea: Assume a model and bootstrap from it.

Parametric Setup

Assume a parametric family:

$$F = F_\theta, \qquad \theta \in \Theta.$$

Estimate θ by:

$$\hat{\theta} = \hat{\theta}(X_1, \ldots, X_n).$$

Bootstrap approximation:

$$\boxed{\mathcal{L}_{F_\theta}(T_n) \approx \mathcal{L}_{F_{\hat{\theta}}}(T_n^*)}$$

Parametric Bootstrap Algorithm

1. Compute $\hat{\theta}$ using MLE.
2. Generate
$$X_1^*, \ldots, X_n^* \overset{i.i.d.}{\sim} F_{\hat{\theta}}.$$
3. Compute $T_n^* = T(X_1^*, \ldots, X_n^*)$.
4. Repeat B times to approximate $\mathcal{L}_{F_{\hat{\theta}}}(T_n^*)$.

True Sampling Behavior

Let $X_1, \ldots, X_n \sim \text{Unif}(0, \theta)$ and

$$T_n = M_n = \max(X_1, \ldots, X_n).$$

Then

$$F_{M_n}(x) = \left(\frac{x}{\theta}\right)^n, \quad 0 \le x \le \theta,$$

and the normalized maximum satisfies

$$n(\theta - M_n) \; \Rightarrow \; \text{Exp}(\theta).$$

Parametric Bootstrap Fix

Assume the correct model Unif$(0, \theta)$.
Estimator:

$$\hat{\theta} = M_n.$$

Bootstrap:

$$X_1^*, \ldots, X_n^* \sim \text{Unif}(0, \hat{\theta}), \qquad M_n^* = \max(X_i^*).$$

Then

$$n(\hat{\theta} - M_n^*) \,\Big|\, X_1, \ldots, X_n$$

has the correct exponential-type spacing behavior near the boundary and has no point mass at zero.

Indicator Functional

$$\theta = T(F) = \mathbf{1}\{\mu > c\}, \qquad \hat{\theta} = T(F_n) = \mathbf{1}\{\bar{X} > c\}.$$

If $\mu = c$, then by CLT

$$\sqrt{n}(\bar{X} - c) \Rightarrow N(0, \sigma^2),$$

so

$$\hat{\theta} = \mathbf{1}\{\bar{X} > c\} \Rightarrow \text{Bernoulli}(1/2).$$

Why NP Bootstrap Fails

It can be shown that :

$$\mathbb{P}^*(\bar{X}^* > c) \to \Phi\left(\frac{\sqrt{n}(\bar{X} - c)}{s_n}\right),$$

If $\bar{X} > c$ (even by $O(n^{-1/2})$), then typically

$$\mathbb{P}^*(\bar{X}^* > c) \to 1,$$

so $\hat{\theta}^* = 1$ with high probability.
Hence the NP bootstrap collapses to a point mass and understates uncertainty near $\mu = c$.
Note: Note that the marginal distribution of bootstrap statistic is Uniform.

Parametric Bootstrap Fix

Assume $X_i \sim N(\mu, \sigma^2)$.
Bootstrap:

$$X_1^*, \ldots, X_n^* \sim N(\bar{X}, s_n^2), \qquad \hat{\theta}^* = \mathbf{1}\{\bar{X}^* > c\}.$$

Then (conditionally)

$$\mathbb{P}^*(\hat{\theta}^* = 1) = \mathbb{P}^*(\bar{X}^* > c) = \Phi\left(\frac{\bar{X} - c}{s_n/\sqrt{n}}\right).$$

If $\mu = c$, then $\frac{\bar{X}-c}{s_n/\sqrt{n}} \to N(0, 1)$ and

$$\mathbb{P}^*(\hat{\theta}^* = 1) \xrightarrow{P} \Phi(Z) = Unif(0, 1),$$

matching the distribution of non-parametric bootstrap. It only helps to reach the distribution without the use of CLT.

Non-smooth Functional

Let $X_i \sim N(\mu, \sigma^2)$ and define

$$\theta = T(F) = \max(0, \mu), \qquad \hat{\theta} = T(F_n) = \max(0, \bar{X}).$$

Note that $T(\cdot)$ is non-differentiable at $\mu = 0$.

Limit Distribution at $\mu = 0$

Suppose $\mu = 0$. Then by the CLT,

$$\sqrt{n}\,\bar{X} \Rightarrow N(0, \sigma^2).$$

Hence

$$\sqrt{n}\,\hat{\theta} = \sqrt{n}\max(0, \bar{X}) = \max\big(0, \sqrt{n}\bar{X}\big) \Rightarrow \max(0, Z),$$

where $Z \sim N(0, \sigma^2)$.
Thus the limit law is a **half-normal distribution** with an atom at 0.

Why NP Bootstrap Fails

The nonparametric bootstrap draws

$$X_1^*, \dots, X_n^* \sim F_n, \qquad \hat{\theta}^* = \max(0, \bar{X}^*).$$

Conditionally on the data,

$$\sqrt{n}(\bar{X}^* - \bar{X}) \Rightarrow N(0, s_n^2),$$

so approximately

$$\bar{X}^* \approx \bar{X} + \frac{s_n}{\sqrt{n}} Z^*, \quad Z^* \sim N(0,1).$$

Hence

$$\hat{\theta}^* = \max\left(0, \bar{X} + \frac{s_n}{\sqrt{n}} Z^*\right).$$

Why NP Bootstrap Fails

At the boundary $\mu = 0$, the sample mean $\bar{X}$ is $O_p(n^{-1/2})$. Define $W_n = \frac{\sqrt{n}(\bar{X}-0)}{s_n}$, where $W_n \xrightarrow{d} N(0,1)$.
The bootstrap statistic can be written as:

$$\hat{\theta}^* = \frac{s_n}{\sqrt{n}} \max(0, W_n + Z^*), \quad Z^* \sim N(0,1)$$

The Discrepancy:

- **True Sampling Distribution:** The distribution of $\sqrt{n}(\hat{\theta} - \theta)$ converges to the distribution of $s \cdot \max(0, Z)$.
- **Bootstrap Distribution:** Conditioned on the data (W_n), the distribution converges to $s \cdot \max(0, w + Z^*)$.

Parametric Bootstrap

Assume $X_i \sim N(\mu, \sigma^2)$.
Parametric bootstrap:

$$X_1^*, \ldots, X_n^* \sim N(\bar{X}, s_n^2), \qquad \hat{\theta}^* = \max(0, \bar{X}^*).$$

Then conditionally,

$$\bar{X}^* \sim N\left(\bar{X}, \frac{s_n^2}{n}\right),$$

so

$$\hat{\theta}^* = \max\left(0, \bar{X} + \frac{s_n}{\sqrt{n}} Z^*\right).$$

Parametric Bootstrap at $\mu = 0$

If $\mu = 0$, then

$$\frac{\bar{X}}{s_n/\sqrt{n}} \Rightarrow N(0,1).$$

Hence

$$\sqrt{n}\,\hat{\theta}^* = \max\left(0, \frac{\bar{X}}{s_n/\sqrt{n}} + Z^*\right)$$

which is same as the non-parametric bootstrap..
Thus the parametric bootstrap also **fails to recover** the correct half-normal limit of $\sqrt{n}\hat{\theta}$.

Conclusion

- Parametric bootstrap fixes the distribution of bootstrap statistics by assuming its distribution.
- So it works well and usually works better when we can make reasonable assumptions about the underlying distribution.
- But in many cases, it fails exactly where the non-parametric bootstrap fails, when the estimator is not "differentiable."
- This motivates alternatives such as:
  - m out of n bootstrap, or
  - delete-d jackknife.

The Problem of Scale

The failure of the standard bootstrap near a "kink" ($\mu = 0$) stems from the **scaling parity**:

- **Sample Variation:** $\bar{X} - \mu = O_p(n^{-1/2})$
- **Bootstrap Variation:** $\bar{X}^* - \bar{X} = O_p(n^{-1/2})$

When $\mu = 0$, these two terms are of the same order. The bootstrap distribution becomes a **random shift** of the true distribution, preventing consistency.

The *m*-out-of-*n* Solution

To restore consistency, we resample m observations **without replacement** (or with replacement) where:

$$m \to \infty \quad \text{and} \quad \frac{m}{n} \to 0$$

Why it works:

- **New Scaling:** $\bar{X}_m^* - \bar{X} = O_p(m^{-1/2})$.
- Since $m \ll n$, the bootstrap variation is **orders of magnitude larger** than the sample variation ($m^{-1/2} \gg n^{-1/2}$).

The "jitter" from the original sample $\bar{X}$ becomes negligible relative to the bootstrap spread, effectively "washing out" the local non-differentiability.

The m-out-of-n Bootstrap Algorithm

Given a sample $\mathcal{X} = \{X_1, \ldots, X_n\}$ and an estimator $\hat{\theta} = T(F_n)$:

1. **Choose Resample Size:** Select m such that $1 \ll m \ll n$ (commonly $m = n^\alpha$ for $0 < \alpha < 1$).
2. **Resample:** Draw m items from $\mathcal{X}$ at random, **without replacement**, to form $\mathcal{X}_m^* = \{X_1^*, \ldots, X_m^*\}$.
3. **Evaluate:** Compute the bootstrap statistic $\hat{\theta}_m^* = T(F_m^*)$.
4. **Repeat:** Perform steps 2–3 B times to obtain the empirical distribution of $\hat{\theta}_m^*$.
5. **Scale:** Use the distribution of $\tau_m(\hat{\theta}_m^* - \hat{\theta})$ to approximate the distribution of $\tau_n(\hat{\theta} - \theta)$, where τ is the convergence rate (e.g., $\sqrt{n}$).

Note: Sampling *without replacement* is often preferred here to ensure the m-out-of-n distribution is well-defined even when the functional is extremely sensitive.

Probability of Picking $X_{(n-k+1)}$

Let $X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}$ be the ordered original sample. Let $\hat{\theta}_m^*$ be the maximum of m bootstrap draws with replacement from $\{X_1, \ldots, X_n\}$. The probability that the bootstrap maximum is **exactly** the k-th largest value ($X_{(n-k+1)}$) is:

$$P^*(\hat{\theta}_m^* = X_{(n-k+1)}) = P^*(\hat{\theta}_m^* \leq X_{(n-k+1)}) - P^*(\hat{\theta}_m^* \leq X_{(n-k)})$$

Since we draw m times with replacement, $P^*(\hat{\theta}_m^* \leq X_{(j)}) = (j/n)^m$:

$$P^*(\hat{\theta}_m^* = X_{(n-k+1)}) = \left(\frac{n-k+1}{n}\right)^m - \left(\frac{n-k}{n}\right)^m$$

The Limit Distribution of Indices

Let $n, m \to \infty$ such that $m/n \to \lambda$. Using $(1 - x/n)^m \to e^{-\lambda x}$:

$$P^*(\hat{\theta}_m^* = X_{(n-k+1)}) \to e^{-\lambda(k-1)} - e^{-\lambda k}$$

The Geometric Structure: Note that these probabilities form a **Geometric distribution** with $p = 1 - e^{-\lambda}$:

$$P^*(k) = (1-p)^{k-1} p$$

The Failure Case ($\lambda > 0$): For the standard bootstrap ($m = n \implies \lambda = 1$), the probability of hitting the maximum ($k = 1$) is $1 - e^{-1} \approx 0.632$. This **point mass** prevents convergence to the continuous true limit.

Consistency via $m/n \to 0$

To recover the continuous limit, we require $m/n \to \lambda = 0$.
The Vanishing Point Mass: As $m/n \to 0$, the "success probability" $p = 1 - e^{-m/n}$ of picking the sample maximum vanishes. Specifically, $p \approx m/n \to 0$.
Spreading the Mass: Since $p \to 0$, the mass of the Geometric distribution K spreads across a large number of order statistics. This "thinned" selection allows the discrete jumps between $X_{(n-k)}$ to be smoothed out in the limit.

Proof of Limit via Gamma Mixture

For Unif$(0, \theta)$, the spacing between the top order statistics $m(\hat{\theta}_n - X_{(n-k+1)})$ converges to a **Gamma distribution** with shape k and scale $\theta/n \cdot n/m = \theta/m$:

$$m(\hat{\theta}_n - X_{(n-k+1)}) \xrightarrow{d} f \cdot \text{Gamma}(k, \theta) \quad \text{(rescaled)}$$

Recall our "Digression" result: A Gamma(k, θ) mixed with $K \sim$ Geom(p) results in an Exp$(\theta/p) = \frac{1}{p} Exp(\theta)$ distribution.
Applying the Mixture: The bootstrap statistic is a mixture of these Gammas with $K \sim$ Geom(p) where $p = 1 - e^{-f} \approx f$ as $f \to 0$.

$$f_{\text{boot}}(x) \to \left(\frac{f}{p} Exp(\theta)\right) = Exp(\theta)$$

Conclusion: The m-out-of-n bootstrap converges to the correct **Exponential** limit because it acts as a geometric mixture of spacings.

Scaling Analysis at the Boundary

Assume $X_i \sim N(\mu, 1)$ and we are at the boundary $\mu = 0$. The sample mean behaves as:

$$\bar{X}_n = \frac{1}{\sqrt{n}} W_n, \quad W_n \sim N(0, 1)$$

In the m-out-of-n bootstrap, the resampled mean $\bar{X}_m^*$ satisfies:

$$\bar{X}_m^* = \bar{X}_n + \frac{1}{\sqrt{m}} Z^*, \quad Z^* \sim N(0, 1)$$

The Key Ratio: We examine the bootstrap statistic at its natural scale $\sqrt{m}$:

$$\sqrt{m}(\hat{\theta}_m^* - \hat{\theta}_n) = \sqrt{m}\left(\max(0, \bar{X}_m^*) - \max(0, \bar{X}_n)\right)$$

Asymptotic Negligibility of Sample Noise

Substitute the expressions for $\bar{X}_n$ and $\bar{X}_m^*$:

$$\sqrt{m}(\hat{\theta}_m^* - \hat{\theta}_n) = \max\left(0, \sqrt{m}\bar{X}_n + Z^*\right) - \max\left(0, \sqrt{m}\bar{X}_n\right)$$

Now, analyze the term $\sqrt{m}\bar{X}_n$ under the condition $m/n \to 0$:

$$\sqrt{m}\bar{X}_n = \sqrt{m}\left(\frac{W_n}{\sqrt{n}}\right) = \sqrt{\frac{m}{n}}W_n$$

Since $W_n = O_p(1)$ and $\frac{m}{n} \to 0$:

$$\sqrt{\frac{m}{n}}W_n \xrightarrow{p} 0$$

Restoration of the Limit Distribution

By Slutsky's Theorem, as $n, m \to \infty$ and $m/n \to 0$, the "jitter" from the sample mean vanishes:

$$\max(0, \text{small} + Z^*) - \max(0, \text{small}) \xrightarrow{d} \max(0, Z^*) - 0$$

Conclusion:

$$\sqrt{m}(\hat{\theta}^*_m - \hat{\theta}_n) \xrightarrow{d} \max(0, Z^*)$$

Final Result:

- The limit is exactly the **Half-Normal** distribution.
- This matches the true limit of $\sqrt{n}(\hat{\theta}_n - 0)$.
- Consistency is restored because the bootstrap spread $(1/\sqrt{m})$ is so much larger than the sample's distance from the boundary $(1/\sqrt{n})$.

The Indicator Functional Failure

Consider the estimator $\hat{\theta}_n = \mathbb{I}(\bar{X}_n > 0)$ where $X_i \sim N(0, 1)$. The true parameter is $\theta = \mathbb{I}(0 > 0) = 0$.

True Sampling Distribution: Since $\bar{X}_n \sim N(0, 1/n)$, the estimator is a Bernoulli random variable:

$$\hat{\theta}_n = \begin{cases} 1 & \text{with probability } 1/2 \\ 0 & \text{with probability } 1/2 \end{cases}$$

The distribution of $\hat{\theta}_n - \theta$ is discrete and does not vanish as $n \to \infty$.

m-out-of-n Convergence

Now let $m \to \infty$ such that $m/n \to 0$. The bootstrap estimate is $\hat{\theta}^*_m = \mathbb{I}(\bar{X}^*_m > 0)$. Using the scaling $\bar{X}^*_m = \bar{X}_n + \frac{1}{\sqrt{m}} Z^*$:

$$P^*(\hat{\theta}^*_m = 1) = P^*\left(\bar{X}_n + \frac{1}{\sqrt{m}} Z^* > 0\right) = P^*(Z^* > -\sqrt{m}\bar{X}_n)$$

The Limit: As derived previously, $\sqrt{m}\bar{X}_n = \sqrt{\frac{m}{n}} W_n \xrightarrow{p} 0$.

$$P^*(\hat{\theta}^*_m = 1) \to P^*(Z^* > 0) = \frac{1}{2}$$

Conclusion: The m-out-of-n bootstrap distribution converges to *Bernoulli*$(1/2)$, which is the **correct true limit distribution**. Consistency is restored.

Fix for jackknife

The standard Jackknife (delete-1) is essentially a linear approximation. It fails for **non-smooth statistics** (e.g., the median, the maximum, or L-statistics) because:

- **Too Little Variation:** Deleting only 1 observation out of n produces replicates $\hat{\theta}_{(i)}$ that are "stiff." The change is too infinitesimal to capture the distribution's curvature.
- **Discretization:** For the median, the jackknife replicates can only take a few discrete values regardless of n, preventing convergence to a continuous limit.

Requirement for Consistency: To handle non-differentiable functionals, the "perturbation" to the data must be more aggressive. We must delete $d \gg 1$ observations.

The Delete-d Logic

To restore consistency, we delete a subset of size d and compute the statistic on the remaining $r = n - d$ observations. We require the "dosage" to satisfy:

$$r \to \infty \quad \text{and} \quad \frac{r}{n} \to 0 \quad (\text{equivalently } d/n \to 1)$$

Why it works:

- **The Subsampling Connection:** r plays the exact same role as m in the m-out-of-n bootstrap.
- **Smoothing:** Deleting a large portion of the data ($d \approx n$) forces the estimator to fluctuate across different order statistics, "washing out" local non-smoothness.

The Delete-d Jackknife Algorithm

Given a sample $\mathcal{X} = \{X_1, \ldots, X_n\}$ and an estimator $\hat{\theta} = T(F_n)$:

1. **Choose Subset Size:** Select $r = n - d$ such that $r \ll n$ (e.g., $r = \sqrt{n}$).
2. **Identify Subsets:** Let $\mathcal{S}$ be the collection of all $\binom{n}{r}$ subsets of size r.
3. **Evaluate:** For each subset $s \in \mathcal{S}$, compute the subset statistic $\hat{\theta}_{r,s}$.
4. **Estimate Bias:**

$$\widehat{\text{Bias}}_{Jd} = \frac{r}{d}(\hat{\theta}_{jd} - \bar{\theta}_r) \text{ where } \hat{\theta}_{jd} = \frac{1}{\binom{n}{d}} \sum_{s \in \mathcal{S}} \hat{\theta}_{r,s}$$

   where $\bar{\theta}_r$ is the average of the $\hat{\theta}_{r,s}$ replicates.

5. **Estimate Variance:**

$$\widehat{\text{Var}}_{Jd} = \frac{r}{d \cdot \binom{n}{d}} \sum_{s \in \mathcal{S}} (\hat{\theta}_{r,s} - \hat{\theta}_{jd})^2$$

Delete-d Jackknife bias estimate

Assume the estimate $\hat{\theta}$ can be written as:

$$\hat{\theta} = \theta + \frac{b}{n} + O\left(\frac{1}{n^2}\right)$$

Then:

$$\hat{\theta}_{r,s} = \theta + \frac{b}{r} + O\left(\frac{1}{r^2}\right)$$

$$\implies \hat{\theta}_{r,s} - \hat{\theta} = \frac{b(n-r)}{nr} + O\left(\frac{1}{r^2}\right)$$

$$\implies \frac{r}{d}(\hat{\theta}_{r,s} - \hat{\theta}) = \frac{b}{n} + O\left(\frac{1}{r(n-r)}\right)$$

$$\implies \widehat{Bias}_{jd} \approx \frac{r}{d}(\hat{\theta}_{jd} - \hat{\theta})$$

So the delete-d jackknife bias correctd estimate can be written as:

$$\hat{\theta}^{bc}_{jd} = \frac{n\hat{\theta} - r\hat{\theta}_{jd}}{d}$$

Delete-d Jackknife variance estimate

If we proceed in a similar manner to delete-1 jackknife, we can define pseudo values as:

$$\theta^*_{r,s} = \frac{n\hat{\theta} - r\hat{\theta}_{r,s}}{d}$$

Observe it acts like the average of d many "corrected" observations for θ_s. So if θ had variance σ^2 then θ^* should have variance σ^2/d, whereas as our original $\hat{\theta}_{r,s}$ used r observations. So we need to apply a correction factor of d/r Using this we can get:

$$\widehat{Var_{jd}}(\hat{\theta}) = \frac{d}{r}\frac{1}{\binom{n}{d}}\sum_{s\in\mathcal{S}}\frac{r^2}{d^2}(\hat{\theta}_{r,s} - \hat{\theta}_{jd})^2 = \frac{r}{d\binom{n}{d}}\sum_{s\in\mathcal{S}}(\hat{\theta}_{r,s} - \hat{\theta}_{jd})^2$$

Example: $(\max(0, \mu))$

Consider the non-smooth functional $\theta = T(\mu) = \max(0, \mu)$ at the point of non-differentiability $\mu = 0$. Assume $X_1, \ldots, X_n \sim N(0, \sigma^2)$.

1. Distribution of Replicates: The subset mean follows $\bar{X}_{r,s} \sim N(0, \sigma^2/r)$. The replicates $\hat{\theta}_{r,s} = \max(0, \bar{X}_{r,s})$ follow a **half-normal distribution**.

2. Moments of Replicates:

- Second Moment: $E[\hat{\theta}^2_{r,s}] = \frac{1}{2}\left(\frac{\sigma^2}{r}\right)$
- First Moment: $E[\hat{\theta}_{r,s}] = \frac{\sigma}{\sqrt{2\pi r}}$
- Replicate Variance: $\mathrm{Var}(\hat{\theta}_{r,s}) = \frac{\sigma^2}{r}\left(\frac{1}{2} - \frac{1}{2\pi}\right) = \frac{\sigma^2}{r}\left(\frac{\pi-1}{2\pi}\right)$

Variance Consistency for $\max(0, \mu)$

3. Jackknife Variance Estimate: Applying the Delete-d scaling factor r/d:

$$\widehat{\text{Var}}_{Jd} = \frac{r}{d} \cdot \text{Var}(\hat{\theta}_{r,s}) = \frac{\sigma^2}{d}\left(\frac{\pi - 1}{2\pi}\right)$$

4. Convergence Analysis: The true variance of the estimator $\hat{\theta} = \max(0, \bar{X}_n)$ is:

$$\text{Var}_{true} = \frac{\sigma^2}{n}\left(\frac{\pi - 1}{2\pi}\right)$$

The ratio of the estimate to the truth is:

$$\frac{\widehat{\text{Var}}_{Jd}}{\text{Var}_{true}} = \frac{n}{d}$$

Conclusion: As $n, d \to \infty$ with $d/n \to 1$, the ratio $\to 1$. The Delete-d Jackknife is perfectly consistent in this case, unlike the Delete-1 version.

Jackknife as a Perturbation

The Delete-1 Jackknife replicate $\hat{\theta}_{(i)}$ can be viewed as the functional $T(\cdot)$ evaluated at a perturbed version of the empirical distribution F_n.
Recall $F_n = \frac{1}{n}\sum_{j=1}^{n} \delta_{X_j}$. If we remove X_i, the remaining $n-1$ points each get weight $\frac{1}{n-1}$. We can write this "deleted" distribution $F_{n-1}^{(i)}$ as:

$$F_{n-1}^{(i)} = \frac{n}{n-1} F_n - \frac{1}{n-1} \delta_{X_i}$$

In the language of influence functions, this is a perturbation of F_n in the direction of δ_{X_i} with a "step size" $\epsilon = -\frac{1}{n-1}$:

$$F_\epsilon = (1-\epsilon)F_n + \epsilon\delta_{X_i} \implies \hat{\theta}_{(i)} = T(F_n + \epsilon(\delta_{X_i} - F_n))$$

The Logic: The standard Jackknife is a **linear approximation** using a discrete step of magnitude $1/(n-1)$.

From Jackknife to Influence Functions

Using the definition of the Influence Function $IF(x; T, F) = \lim_{\epsilon \to 0} \frac{T(F_\epsilon) - T(F)}{\epsilon}$, and substituting our Jackknife $\epsilon = -\frac{1}{n-1}$:

$$\psi(X_i) = IF(X_i; T, F_n) \approx \frac{T(F_{n-1}^{(i)}) - T(F_n)}{-1/(n-1)} = (n-1)(\hat{\theta} - \hat{\theta}_{(i)})$$

This identifies the **Jackknife Replicates** as empirical approximations of the influence values.

Note: This can be used to explain jackknife bias estimate.

Connection to Variance:

Recall the Jackknife variance estimate:

$$
\begin{aligned}
\widehat{\text{Var}}_{J1} &= \frac{n-1}{n}\sum_{i=1}^{n}(\hat{\theta}_{(i)} - \bar{\theta})^2 \\
&= \frac{n-1}{n}\sum_{i=1}^{n}(\hat{\theta}_{(i)} - \hat{\theta})^2 + (n-1)(\hat{\theta} - \bar{\theta})^2 \\
&= \frac{1}{n(n-1)}\sum_{i=1}^{n}\psi(X_i)^2 + \frac{((n-1)(\hat{\theta} - \bar{\theta}))^2}{(n-1)} \\
&\approx \frac{1}{n^2}\sum_{i=1}^{n}\psi(X_i)^2
\end{aligned}
$$

We ignore the second term as the numerator goes to squared bias, which is usually $O(1/n^2)$, so the second term becomes $O(1/n^3)$. As $n \to \infty$, this discrete sum of squared differences converges to the variance of the Influence Function.

The Infinitesimal Jackknife (IJ)

The IJ is the **limiting case** of the Jackknife where the perturbation $\epsilon \to 0$. Instead of removing an observation, we compute the directional derivative. **The IJ Variance Estimator:** Using the Empirical Influence Function (EIF) values $\hat{\psi}(X_i)$, the IJ variance is:

$$\widehat{\text{Var}}_{IJ} = \frac{1}{n^2} \sum_{i=1}^{n} \hat{\psi}(X_i)^2$$

Why use IJ?

- **Computational Efficiency:** You compute the statistic **once** and use gradients/derivatives. No re-sampling or n re-evaluations needed.
- **Modern Scaling:** It is the "gold standard" for uncertainty in **Random Forests** and Bagging, where retraining n times is impossible.

Example: Sample Variance

We know for sample variance $\psi(X_i) = (X_i - \mu)^2 - \sigma^2$, so we can take $\hat{\psi}(X_i) = (X_i - \bar{X})^2 - \hat{\mu}_2$. So :

$$\begin{aligned}\widehat{Var_{IJ}} &= \frac{1}{n^2}\sum_{i=1}^{n}\hat{\psi}(X_i)^2 = \frac{1}{n^2}\sum_{i=1}^{n}((X_i - \bar{X})^2 - \hat{\mu}_2)^2 \\ &= \frac{1}{n}\left(\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^4 - \hat{\mu}_2^2\right) = \frac{1}{n}(\hat{\mu}_4 - \hat{\mu}_2^2)\end{aligned}$$

Also:

$$\widehat{Var_{J1}} = \frac{n^2}{(n-1)^3}(\hat{\mu}_4 - \hat{\mu}_2^2)$$

Using these two we have :

$$\frac{\widehat{Var_{J1}}}{\widehat{Var_{IJ}}} = \frac{n^3}{(n-1)^3} \to 1$$

Linear Regression Model

We observe:

$$(X_i, Y_i), \quad i = 1, \ldots, n$$

Model:

$$Y_i = X_i^\top \beta + \varepsilon_i$$

where:

- $X_i \in \mathbb{R}^p$
- β unknown
- ε_i are errors

Matrix Definitions:

$$\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} X_1^\top \\ \vdots \\ X_n^\top \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}$$

OLS estimator:

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y$$

Why Bootstrap in Regression?

Classical theory assumes:

- i.i.d. errors
- homoskedasticity
- often normality

In practice:

- heteroskedastic errors
- non-normal errors
- small samples

Bootstrap provides a data-driven approximation to:

$$\mathcal{L}(\hat{\beta})$$

Case Resampling Bootstrap

Resample entire pairs:

$$(X_i, Y_i)$$

Bootstrap sample:

$$(X_1^*, Y_1^*), \ldots, (X_n^*, Y_n^*)$$

Compute:

$$\hat{\beta}^* = (X^{*\top} X^*)^{-1} X^{*\top} Y^*$$

Repeat B times.
Approximate:

$$\mathcal{L}(\hat{\beta}) \approx \mathcal{L}^*(\hat{\beta}^*)$$

Residual Bootstrap

Fit model once:

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y$$

Compute residuals:

$$\hat{\varepsilon}_i = Y_i - X_i^\top \hat{\beta}$$

Resample residuals:

$$\varepsilon_i^* \sim \{\hat{\varepsilon}_1, \ldots, \hat{\varepsilon}_n\}$$

Generate:

$$Y_i^* = X_i^\top \hat{\beta} + \varepsilon_i^*$$

Refit:

$$\hat{\beta}^* = (X^\top X)^{-1} X^\top Y^*$$

Case vs Residual Bootstrap

Case Bootstrap:

- Treats (X_i, Y_i) as i.i.d.
- No model assumption on errors
- Robust to heteroskedasticity

Residual Bootstrap:

- Assumes linear model correct
- Design matrix fixed
- Requires homoskedastic errors

Asymptotic Linearity of OLS

Rewrite model:

$$Y_i = X_i^\top \beta + \varepsilon_i$$

also from OLS we know:

$$\sum_{i=1}^{n} X_i (Y_i - X_i^T \hat{\beta}) = 0$$

Multiplying the first equation by X_i^T and summing w.r.t i, then subtracting the 2nd equation gets us:

$$\frac{1}{n}(X^\top X)(\hat{\beta} - \beta) = \frac{1}{n}\sum_{i=1}^{n} X_i^\top \varepsilon_i$$

$$\implies \hat{\beta} - \beta = \frac{1}{n}\sum_{i=1}^{n} \hat{A}^{-1} X_i (Y_i - X_i^T \beta)$$

Here $\hat{A} = \frac{1}{n}(X^\top X)$.

Influence Function for Regression

Define functional:

$$\beta(F) = \arg\min_{\beta} \int (y - x^\top \beta)^2 dF(x, y)$$

So β satisfies $\int x(y - x^\top \beta) dF(x, y) = 0$.
And β_ϵ satisfies $(1 - \epsilon) \int x(y - x^\top \beta_\epsilon) dF(x, y) + \epsilon x_0 (y_0 - x_0^\top \beta_\epsilon) = 0$.
Differentiate w.r.t ϵ and evaluate at $\epsilon = 0$ to get the influence function:

$$IF(x, y; \beta, F) = A^{-1} x (y - x^\top \beta)$$

where:

$$A = E[XX^\top]$$

Thus:

$$\hat{\beta} - \beta = \frac{1}{n} \sum IF(X_i, Y_i) + o_p(n^{-1/2})$$

OLS is asymptotically linear.

Heteroskedastic Errors

If:

$$Var(\varepsilon_i \mid X_i) \neq \sigma^2$$

Residual bootstrap may fail.
Better choices:

- Case bootstrap
- Wild bootstrap

Bootstrap validity depends on correct structural assumptions.

The Wild Bootstrap: Motivation

The standard **residual bootstrap** assumes ε_i are i.i.d., which implies $Var(\varepsilon_i|X_i) = \sigma^2$. If heteroskedasticity is present:

- The empirical distribution of centered residuals $\hat{\varepsilon}$ fails to capture the conditional variance $Var(\varepsilon_i|X_i)$.
- **Wild Bootstrap** solves this by resampling each residual *locally*.

The key idea is to generate bootstrap errors:

$$\varepsilon_i^* = \hat{\varepsilon}_i \cdot v_i$$

where v_i is an independent random variable with $E[v_i] = 0$ and $E[v_i^2] = 1$.

The Wild Bootstrap: Algorithm

1. Fit the OLS model $Y = X\beta + \varepsilon$ to get $\hat{\beta}$ and residuals $\hat{\varepsilon}_i = Y_i - X_i^\top \hat{\beta}$.
2. For each bootstrap replication $b = 1, \ldots, B$:
    - Generate n auxillary random variables $v_{i,b}$ (e.g., Rademacher).
    - Form bootstrap response: $Y_{i,b}^* = X_i^\top \hat{\beta} + \hat{\varepsilon}_i v_{i,b}$.
    - Calculate bootstrap OLS estimate:

$$\hat{\beta}_b^* = (X^\top X)^{-1} X^\top Y_b^*$$

3. Use the distribution of $(\hat{\beta}_b^* - \hat{\beta})$ to estimate the distribution of $(\hat{\beta} - \beta)$.

Note: Unlike the Case Bootstrap, the regressors X_i are treated as *fixed* in the Wild Bootstrap.

Choosing the Weight Distribution v_i

The choice of the distribution for v_i affects the refinement of the bootstrap. Common choices include:

Rademacher Distribution (Recommended)

$$v_i = \begin{cases} 1 & \text{with prob } 0.5 \\ -1 & \text{with prob } 0.5 \end{cases}$$

It is often preferred because $v_i^2 = 1$ always, perfectly preserving the squared residual.

Mammen's Distribution

Designed to match the 3rd moment ($E[v_i^3] = 1$):

$$v_i = \begin{cases} -(\sqrt{5}-1)/2 & \text{with prob } (\sqrt{5}+1)/(2\sqrt{5}) \\ (\sqrt{5}+1)/2 & \text{with prob } (\sqrt{5}-1)/(2\sqrt{5}) \end{cases}$$

Why it Works: Moment Matching

Conditional on the data, the bootstrap error ε_i^* satisfies:

- $E^*[\varepsilon_i^*] = \hat{\varepsilon}_i E[v_i] = 0$
- $Var^*(\varepsilon_i^*) = \hat{\varepsilon}_i^2 E[v_i^2] = \hat{\varepsilon}_i^2$

Recall the Influence Function from earlier:

$$\hat{\beta} - \beta \approx \frac{1}{n} \sum A^{-1} X_i \varepsilon_i$$

As ϵ_i^* has the variance, ϵ_i^2 it recovers the proper variance of $\hat{\beta}$.

Bootstrap Confidence Intervals

Key Results

$$\sqrt{n}(\hat{\theta}^* - \hat{\theta}) \sim N(0, \sigma^2) \quad \text{(Bootstrap CLT)}$$

$$\sqrt{n}(\hat{\theta} - \theta) \sim N(0, \sigma^2) \quad \text{(Normal CLT)}$$

Idea

Replace the unknown sampling distribution of $\hat{\theta} - \theta$ by the empirical distribution of $\hat{\theta}^* - \hat{\theta}$.

Bootstrap t-Interval

Suppose we observe:

$$X_1, \ldots, X_n$$

1. Generate bootstrap samples

$$\{X_1^{*(b)}, \ldots, X_n^{*(b)}\}$$

2. Compute $\hat{\theta}^{*(b)}$
3. Compute $\widehat{se}^{*(b)}$

Standard Error Estimation

Important:

$\widehat{se}^{*(b)}$ is computed from the b-th sample only

Example: if $\hat{\theta} = \bar{X}$,

$$\widehat{se}^{*(b)} = \sqrt{\frac{1}{n-1}\sum(X_i^{*(b)} - \bar{X}^{*(b)})^2}$$

Standard Error Estimation

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Example: if $\hat{\theta} = \bar{X}$,

$$\widehat{se}^{*(b)} = \sqrt{\frac{1}{n-1}\sum(X_i^{*(b)} - \bar{X}^{*(b)})^2}$$

If no closed form is available:

- Bootstrap inside bootstrap
- Or use Delta method

Delta Method Reminder

If

$$\theta = g(\mu)$$

then

$$Var(\hat{\theta}) \approx (g'(\mu))^2 Var(\hat{\mu})$$

Bootstrap t Statistic

Define

$$Z^{*(b)} = \frac{\hat{\theta}^{*(b)} - \hat{\theta}}{\widehat{se}^{*(b)}}$$

These behave like t-observations.

Bootstrap t Statistic

Define

$$Z^{*(b)} = \frac{\hat{\theta}^{*(b)} - \hat{\theta}}{\widehat{se}^{*(b)}}$$

These behave like t-observations.
Find empirical quantiles:

$$t^*_{\alpha/2}, \quad t^*_{1-\alpha/2}$$

Bootstrap t Confidence Interval:

$$\left(\hat{\theta} - t^*_{1-\alpha/2}\widehat{se}, \quad \hat{\theta} - t^*_{\alpha/2}\widehat{se}\right)$$

Why Go Through This Trouble?

In real problems, the distribution of

$$\hat{\theta} - \theta$$

may **not** be a t-distribution.

Why Go Through This Trouble?

In real problems, the distribution of

$$\hat{\theta} - \theta$$

may **not** be a t-distribution.
Bootstrap captures:

- Skewness
- Kurtosis
- Other non-normal features

Limitations :

- Erratic for very small samples
- May violate parameter range

Example: Estimate of probability p must lie in $(0, 1)$.

Fixing the Range: Transformation

Strategy:

1. Transform parameter
2. Construct CI in transformed scale
3. Transform back

Examples

Bernoulli Probability :
If $\theta = p \in (0, 1)$,

$$\beta = \log \frac{p}{1 - p}$$

Then $\beta \in (-\infty, \infty)$.
Back-transform:

$$p = \frac{e^{\beta}}{1 + e^{\beta}}$$

Example: Correlation
If $\theta = \rho \in (-1, 1)$,

$$\beta = \log \frac{1 - \rho}{1 + \rho}$$

Often yields shorter, better intervals.

Variance Stabilization

If

$$Var(\hat{\theta}) = \sigma^2(\theta)$$

Choose g such that

$$(g'(\theta))^2\sigma^2(\theta) = C$$

$$g(\theta) = \int \frac{1}{\sigma(\theta)} d\theta$$

Example: Bernoulli Mean

$$Var(\bar{X}) = \frac{p(1-p)}{n}$$

Ignore $1/n$:

$$g(p) = \int \frac{1}{\sqrt{p(1-p)}} dp$$

$$g(p) = 2 \arcsin(\sqrt{p})$$

Logit is another option. **Example: Correlation**

$$Var(\hat{\rho}) \approx \frac{(1-\rho^2)^2}{n}$$

$$g(\rho) = \frac{1}{2} \log \frac{1+\rho}{1-\rho}$$

General Variance Stabilization

If explicit formula unavailable:

1. Plot $(\hat{\theta}^{*(b)}, \widehat{se}^{*(b)})$
2. Fit smooth function $\hat{s}(u)$
3. Define

$$g(x) = \int^{x} \frac{1}{\hat{s}(u)} du$$

Then bootstrap using g.

BC_a Interval

BC_a = **Bias-Corrected and Accelerated** interval.
It improves the percentile interval by adjusting for:

- **Bias** (median bias of bootstrap distribution)
- **Skewness** (change of standard error with parameter)

Instead of using raw percentile levels, we adjust the probability levels before taking quantiles.

Model Motivation

Suppose

$$\phi = g(\theta)$$

Assume the approximation:

$$\frac{\hat{\phi} - \phi}{1 + a\phi} \approx N(-z_0, 1)$$

where:

- z_0 = bias-correction term
- a = acceleration parameter

This is the working model behind BC_a.

Rewriting the Model

Rewrite as:

$$z_0 + \frac{\hat{\phi} - \phi}{1 + a\phi} \sim N(0, 1)$$

Let

$$Z = z_0 + \frac{\hat{\phi} - \phi}{1 + a\phi}$$

Since normal distribution is symmetric:

$$P(Z \leq z_\alpha) = \alpha$$

Deriving the Lower Confidence Limit

From

$$P\left(z_0 + \frac{\hat{\phi} - \phi}{1 + a\phi} \leq z_\alpha\right) = \alpha$$

After algebra:

$$P\left(\phi < \frac{\hat{\phi} + K}{1 - aK}\right) = \alpha$$

where

$$K = z_0 + z_\alpha$$

Thus lower limit:

$$\boxed{\frac{\hat{\phi} + K}{1 - aK}}$$

Upper Limit and Condition

Upper limit obtained similarly using

$$K = z_0 + z_{1-\alpha}$$

Important Condition:

$$1 - aK > 0$$

Otherwise the transformation breaks down.

Estimating Adjusted Quantile Level

We need adjusted probability level α_1.
Suppose

$$P^*(\phi^* < \hat{\phi}(\alpha_1)) = \alpha_1$$

Using BC$_a$ model:

$$P\left(\frac{\phi^* - \hat{\phi}}{1 + a\hat{\phi}} < \frac{K}{1 - aK}\right) = \alpha_1$$

where

$$K = z_0 + z_{\alpha_1}$$

Using the Bootstrap Pivot

Recall

$$Z^* = z_0 + \frac{\phi^* - \hat{\phi}}{1 + a\hat{\phi}} \sim N(0, 1)$$

Thus,

$$P\left(Z^* < z_0 + \frac{K}{1 - aK}\right) = \alpha_1$$

Final Equation for α_1

Therefore,

$$\Phi\left(z_0 + \frac{z_0 + z_{\alpha_1}}{1 - a(z_0 + z_{\alpha_1})}\right) = \alpha_1$$

We solve this to obtain α_1.
Then take bootstrap quantile:

$$\frac{1}{B}\#\{\phi^{*(b)} < \hat{\phi}_{(\alpha_1)}\} = \alpha_1$$

Upper Adjusted Level

Similarly, upper level α_2 satisfies:

$$\Phi\left(z_0 + \frac{z_0 + z_{1-\alpha}}{1 - a(z_0 + z_{1-\alpha})}\right) = \alpha_2$$

BC_a interval:

$$\left(\hat{\phi}_{(\alpha_1)}, \hat{\phi}_{(\alpha_2)}\right)$$

Estimating Bias-Correction z_0

$$\hat{z}_0 = \Phi^{-1}\left(\frac{\#\{\hat{\theta}^{*(b)} < \hat{\theta}\}}{B}\right)$$

Interpretation:
If bootstrap distribution is symmetric around $\hat{\theta}$,

$$\hat{z}_0 \approx 0$$

Estimating Acceleration a (Jackknife)

Let $\hat{\theta}_{(i)}$ be leave-one-out estimate.
Define

$$\bar{\theta}_{(\cdot)} = \frac{1}{n}\sum_{i=1}^{n}\hat{\theta}_{(i)}$$

Then

$$\hat{a} = \frac{\sum(\bar{\theta}_{(\cdot)} - \hat{\theta}_{(i)})^3}{6\left[\sum(\bar{\theta}_{(\cdot)} - \hat{\theta}_{(i)})^2\right]^{3/2}}$$

Accounts for skewness of influence function.

Testing $H_0 : \mu_1 = \mu_2$

Test statistic:

$$t(\mathbf{x}) = \bar{X} - \bar{Y}$$

We approximate its sampling distribution using bootstrap.

Bootstrap Test: Setup

Suppose

$$X_1, \ldots, X_m \quad \text{and} \quad Y_1, \ldots, Y_n$$

Let $N = m + n$.

Under H_0, pool the data:

$$\{X_1, \ldots, X_m, Y_1, \ldots, Y_n\}$$

Bootstrap Test: Procedure

1. Draw bootstrap sample of size N:

$$S^* = \{X_1^*, \dots, X_{m+n}^*\}$$

2. First $m \to S_X^*$

$$S_X^* = \{X_1^*, \dots, X_m^*\}$$

   Remaining $n \to S_Y^*$

$$S_Y^* = \{X_{m+1}^*, \dots, X_{m+n}^*\}$$

Bootstrap Test Statistic

Compute

$$\bar{X}^*, \quad \bar{Y}^*$$

$$t^{*(b)} = \bar{X}^* - \bar{Y}^*$$

Observed statistic:

$$t_{\text{obs}} = \bar{X} - \bar{Y}$$

Achieved Significance Level

$$\widehat{\text{ASL}}_{\text{boot}} = \frac{\#\{t^{*(b)} \geq t_{\text{obs}}\}}{B}$$

Reject H_0 if

$$\widehat{\text{ASL}}_{\text{boot}} < \alpha$$

Permutation Test

Instead of resampling with replacement:
Permute

$$(X_1, \ldots, X_m, Y_1, \ldots, Y_n)$$

Total permutations:

$$\binom{m+n}{m}$$

Permutation ASL

For each permutation compute:

$$t^{*(b)} = \bar{X}^* - \bar{Y}^*$$

$$\widehat{\text{ASL}}_{\text{perm}} = \frac{\#\{t^{*(b)} \geq t_{\text{obs}}\}}{\binom{m+n}{m}}$$

Remark:
Choosing m observations determines the Y group automatically.

Testing via Confidence Intervals

Construct a $(1-\alpha)$ bootstrap confidence interval.

Reject H_0 if the hypothesized value does not lie in the interval.

Standardization Idea

Classical statistic (equal variance):

$$\frac{\bar{X} - \bar{Y}}{\hat{\sigma}\sqrt{\frac{1}{m} + \frac{1}{n}}}$$

Unequal variances:

$$\frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\hat{\sigma}_1^2}{m} + \frac{\hat{\sigma}_2^2}{n}}}$$

We can bootstrap this studentized version.

Bootstrap Under H_0

Assume H_0 true.
Pooled mean:

$$\hat{\mu} = \frac{m\bar{X} + n\bar{Y}}{m + n}$$

Center samples:

$$\tilde{X}_i = X_i - \bar{X} + \hat{\mu}$$

$$\tilde{Y}_i = Y_i - \bar{Y} + \hat{\mu}$$

Studentized Bootstrap Statistic

Bootstrap from centered samples.
Compute

$$\bar{X}^*, \bar{Y}^*$$

$$\hat{\sigma}_1^{*2}, \quad \hat{\sigma}_2^{*2}$$

$$t^{*(b)} = \frac{\bar{X}^* - \bar{Y}^*}{\sqrt{\frac{\hat{\sigma}_1^{*2}}{m} + \frac{\hat{\sigma}_2^{*2}}{n}}}$$

Studentized ASL

Observed statistic:

$$t_{\text{obs}} = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\hat{\sigma}_1^2}{m} + \frac{\hat{\sigma}_2^2}{n}}}$$

$$\widehat{\text{ASL}}_{\text{boot}} = \frac{\#\{t^{*(b)} \geq t_{\text{obs}}\}}{B}$$

Reject if $< \alpha$.

One-Sample Bootstrap Test

Studentized statistic:

$$t^{*(b)} = \frac{\bar{X}^* - \mu_0}{\widehat{se}(\bar{X}^*)}$$

Permutation generally not applicable.
Bootstrap useful when parametric test unavailable.

Kernel Density Estimation

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^{n} \phi\left(\frac{x - X_i}{h}\right)$$

Interpretation:
Add small normal density at each data point.

Role of Bandwidth h

- Small $h \rightarrow$ many sharp peaks
- Larger $h \rightarrow$ peaks merge
- Large enough $h \rightarrow$ unimodal

We find smallest h making density unimodal.

Bootstrap Test for Unimodality

Test:

$$H_0: \text{ Distribution is unimodal}$$

Compute critical bandwidth $\hat{h}$.

$$\widehat{\text{ASL}} = \frac{\#\{\hat{h}^{*(b)} \geq \hat{h}\}}{B}$$

Rescaling for Bootstrap

Kernel estimator has larger variance than true density.
Use smoothed version $\hat{g}(x; \hat{h})$ before bootstrapping.

Bootstrap samples:

$$X_i^* = \bar{Y}^* + \left(1 + \frac{\hat{h}^2}{\hat{\sigma}^2}\right)^{1/2} \left(Y_i^* - \bar{Y}^* + \hat{h}\varepsilon_i\right)$$

$$\varepsilon_i \sim N(0, 1)$$

Unimodality Decision Rule

Compute $\hat{h}^*$ for each bootstrap sample.

$$\widehat{\text{ASL}}_{\text{boot}} = \frac{\#\{\hat{h}^{*(b)} \geq \hat{h}\}}{B}$$

Reject H_0 if

$$\widehat{\text{ASL}}_{\text{boot}} < \alpha$$

Bayesian vs Frequentist Approach

Bayesian Paradigm

Data X
Model: $X \sim f_\theta$
θ is a parameter of interest
Not fixed; it has a distribution

Frequentist Model

Data X
Model: $X \sim P_\theta, \ \theta \in \Theta$
θ is an unknown state of nature
Parameter is fixed but unknown

- Bayesian analysis treats θ as a random variable.
- Frequentist analysis treats θ as a fixed constant.

Prior Knowledge in Bayesian Analysis

Before observing any data, we specify a **prior distribution** for θ.

- Represents prior knowledge or belief about θ
- Encodes information available before observing data

After observing data, this prior information is updated using the likelihood of the data.

Likelihood and Observed Data

When data are observed, they come from the conditional distribution

$$f(x \mid \theta) \quad \text{or} \quad f_\theta(x)$$

- This is true in both Bayesian and frequentist approaches
- Once data are observed, there is no uncertainty about the data itself
- The uncertainty concerns the parameter θ

Posterior Distribution

In Bayesian inference, uncertainty about θ after observing data is represented by the **posterior distribution**.

The posterior distribution is the conditional distribution of θ given the observed data.

Using Bayes' rule:

$$\pi(\theta \mid x) = \frac{f(x \mid \theta)\pi(\theta)}{\int f(x \mid \theta)\pi(\theta)\, d\theta}$$

Posterior Proportionality

The posterior distribution can be written in proportional form:

$$\pi(\theta \mid x) \propto f(x \mid \theta)\pi(\theta)$$

- $f(x \mid \theta)$ is the likelihood
- $\pi(\theta)$ is the prior distribution
- $\pi(\theta \mid x)$ is the posterior distribution

Thus,

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

Example: Normal Likelihood with Normal Prior

Suppose

$$X \sim N(\mu, \sigma^2), \qquad \sigma^2 \text{ known}$$

Prior distribution:

$$\mu \sim N(\theta, \tau^2)$$

n **observed sample points**

$$f(x|\mu) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2\right)$$

Posterior Distribution

Posterior density:

$$\pi(\mu|x) \propto f(x|\mu)\pi(\mu)$$

$$\pi(\mu|x) \propto \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2 - \frac{1}{2\tau^2}(\mu-\theta)^2\right)$$

We now simplify the quadratic expression in μ.

Expanding the Quadratic Terms

Rewrite the likelihood part:

$$\sum_{i=1}^{n}(x_i - \mu)^2 = \sum_{i=1}^{n}(x_i - \bar{x})^2 + n(\mu - \bar{x})^2$$

Thus,

$$\pi(\mu|x) \propto \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \bar{x})^2 - \frac{n}{2\sigma^2}(\mu - \bar{x})^2 - \frac{1}{2\tau^2}(\mu - \theta)^2\right)$$

Ignoring Constant Terms

Terms not involving μ can be ignored.

$$\pi(\mu|x) \propto \exp\left(-\frac{n}{2\sigma^2}(\mu - \bar{x})^2 - \frac{1}{2\tau^2}(\mu - \theta)^2\right)$$

Define

$$k_1 = \frac{n}{2\sigma^2}, \qquad k_2 = \frac{1}{2\tau^2}$$

Quadratic Simplification

Let

$$a = \mu - \bar{x}, \qquad b = \mu - \theta$$

Then

$$k_1 a^2 + k_2 b^2 = \frac{(k_1 a + k_2 b)^2}{k_1 + k_2} + \frac{k_1 k_2}{k_1 + k_2}(a - b)^2$$

Note

$$(a - b)^2 = (\bar{x} - \theta)^2$$

which does not involve μ.

Posterior Distribution

Therefore

$$\pi(\mu|x) \propto \exp\left(-\frac{1}{k_1+k_2}\left((k_1+k_2)\mu - k_1\bar{x} - k_2\theta\right)^2\right)$$

Hence the posterior distribution is

$$\mu|x \sim N(\mu_p, \sigma_p^2)$$

Posterior Mean

Posterior mean:

$$\mu_p = \frac{k_1\bar{x} + k_2\theta}{k_1 + k_2}$$

Substituting k_1, k_2:

$$\mu_p = \frac{\frac{n\bar{x}}{\sigma^2} + \frac{\theta}{\tau^2}}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}$$

- Weighted average of sample mean and prior mean
- Weights depend on variances

Posterior Variance

Posterior variance:

$$\sigma_p^2 = \frac{1}{2(k_1 + k_2)} = \frac{1}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}$$

Observation

- As $n \to \infty$, prior influence disappears
- Posterior converges to the frequentist estimator distribution

Example 2: Binomial with Beta Prior

Likelihood:

$$f(x|p) = \binom{n}{x} p^x (1-p)^{n-x}$$

For N samples,

$$f(x|p) \propto p^{\sum x_i} (1-p)^{nN - \sum x_i}$$

Prior:

$$\pi(p) \propto p^{\alpha-1} (1-p)^{\beta-1}$$

Posterior Distribution

Combining likelihood and prior,

$$\pi(p|x) \propto p^{\sum x_i + \alpha - 1}(1-p)^{nN - \sum x_i + \beta - 1}$$

Therefore

$$p|x \sim \text{Beta}(\alpha + \sum x_i,\ \beta + nN - \sum x_i)$$

Posterior Mean

For a Beta(α, β) distribution

$$E[p] = \frac{\alpha}{\alpha + \beta}$$

$$Var(p) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

Posterior mean:

$$E[p|x] = \frac{\alpha + \sum x_i}{\alpha + \beta + nN}$$

Posterior Variance

Posterior variance:

$$Var(p|x) = \frac{(\alpha + \sum x_i)(\beta + nN - \sum x_i)}{(\alpha + \beta + nN)^2(\alpha + \beta + nN + 1)}$$

- Posterior combines prior information and observed data
- Influence of the prior decreases as sample size increases

Improper and Non-informative Priors

In many posterior calculations we write

$$\pi(\theta|x) \propto f(x|\theta)\pi(\theta)$$

- Constants are often ignored while computing posterior densities
- Therefore the prior does not always need to be a proper density
- This leads to the concept of **improper priors**

Improper Priors

Suppose we have no prior knowledge about a parameter.

A natural choice is a uniform prior over the entire real line:

$$\pi(\theta) = c, \qquad \forall\theta \in \mathbb{R}$$

However,

$$\int_{-\infty}^{\infty} \pi(\theta)\, d\theta = \infty$$

Hence this is **not a valid probability density**.

Definition: Improper Prior

A prior is called **improper** if

$$\int \pi(\theta)\, d\theta = \infty$$

If instead

$$\int \pi(\theta)\, d\theta = C < \infty$$

then the prior is called an **unnormalized prior**.

Improper priors can still be used if the resulting posterior distribution is proper.

Example

Consider the improper prior

$$\pi(\theta) = c$$

Then the posterior becomes

$$\pi(\theta|x) \propto f(x|\theta)\pi(\theta)$$

$$\pi(\theta|x) \propto f(x|\theta)$$

Thus the posterior is proportional to the likelihood.

Non-informative Priors

In this example, using the prior

$$\pi(\theta) = c$$

does not change the shape of the posterior distribution.

- The posterior depends only on the likelihood
- No additional information is introduced by the prior

Such priors are called **non-informative priors**.

Point and Interval Estimation

Loss functions measure the performance of estimators.

One commonly used loss function is the L^2 loss:

$$L(\theta, T(x)) = (T(x) - \theta)^2$$

Risk of an estimator

$$E_\theta[L(\theta, T(x))] = R(\theta, T)$$

Bayes risk

$$E_\pi[L(\theta, T(x))] = R_\pi$$

Posterior Expected Loss

Since the sample has already been observed, the best estimator is the one that minimizes the expected loss with respect to the posterior distribution.

Average squared error:

$$E[(\theta - T(x))^2]$$

In the Bayesian framework we minimize

$$\int (\theta - T(x))^2 \, \pi(\theta \mid x) \, d\theta$$

The estimator $T(x)$ is chosen to minimize this posterior expected loss.

Bayes Estimator under Squared Loss

It can be shown that the value minimizing

$$\int (\theta - T(x))^2 \, \pi(\theta \mid x) \, d\theta$$

is

$$T(x) = E[\theta \mid x]$$

- Under squared error loss, the Bayes estimator is the **posterior mean**.

Other Loss Functions

Consider the absolute loss function

$$L(\theta, T(x)) = |\theta - T(x)|$$

The Bayes estimator becomes

$$T(x) = \text{posterior median}$$

Thus different loss functions lead to different Bayes estimators.

Mode-based Loss Function

Consider the loss function

$$L(\theta, T(x)) = \mathbf{1}(|\theta - T(x)| < k)$$

This chooses the center of the interval of length $2k$ having the highest posterior probability.

As $k \to 0$, the estimator approaches the

posterior mode or Maximum A Posteriori (MAP) estimato

Discrete Case

For a discrete posterior distribution,

$$L(\theta, T(x)) = \mathbf{1}(\theta = T(x))$$

The estimator maximizing posterior probability is obtained.

This is called the

Maximum A Posteriori (MAP) estimator

Example: Normal-Normal Model

Suppose

$$X \sim N(\mu, \sigma^2), \qquad \sigma^2 \text{ known}$$

Prior:

$$\mu \sim N(\theta, \tau^2)$$

Posterior mean:

$$E(\mu \mid x) = \frac{\frac{n\bar{x}}{\sigma^2} + \frac{\theta}{\tau^2}}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}$$

Posterior Variance

Posterior variance:

$$\mathrm{Var}(\mu \mid x) = \frac{1}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}$$

Since the posterior distribution is symmetric and unimodal,

$$E(\mu|x)$$

is the Bayes estimator under the above loss functions.

Frequentist Confidence Intervals

In the frequentist approach,

- θ is not treated as a random variable
- We cannot say $\theta \in (a, b)$ with probability $1 - \alpha$

Instead,

$$P(a(X) < \theta < b(X)) = 1 - \alpha$$

refers to the long-run coverage probability of the procedure.

Bayesian Credible Intervals

In the Bayesian framework

- θ is treated as a random variable
- The data are already observed

Therefore it is meaningful to say

$$P(a < \theta < b \mid x) = 1 - \alpha$$

This interval is called a

credible interval

Credible Interval

For $0 < \alpha < 1$, a $(1-\alpha)$ level credible interval for θ is an interval (a, b) such that

$$P(\theta \in (a, b) \mid x) = 1 - \alpha$$

- θ is treated as a random variable
- The probability statement is conditional on the observed data

Thus the interval contains θ with posterior probability $1 - \alpha$.

Choosing a Credible Interval

There are many intervals satisfying

$$P(\theta \in (a, b) \mid x) = 1 - \alpha$$

- The interval can be shifted
- The shape of the interval may vary

Therefore it is desirable to choose the interval with the **smallest possible length**.

Highest Posterior Density (HPD) Interval

The interval with the smallest length is called the

Highest Posterior Density (HPD) interval

It is defined as

$$(L, U) = \{\theta \in \Theta : \pi(\theta \mid x) > k(x)\}$$

where $k(x)$ is chosen such that

$$P(\theta \in (L, U) \mid x) = 1 - \alpha$$

Properties of HPD Intervals

- Every point inside the interval has higher posterior density than any point outside the interval
- Among all credible intervals with probability $1 - \alpha$, the HPD interval has the shortest length
- If the posterior distribution is symmetric and unimodal, the HPD interval coincides with the equal–tail credible interval

Example: Normal Model

Normal likelihood with Normal prior

$$X \sim N(\mu, \sigma^2), \qquad \mu \sim N(\theta, \tau^2)$$

Posterior distribution:

$$\pi(\mu \mid x) \sim N(\hat{\mu}, \hat{\sigma}^2)$$

Posterior Parameters

Posterior mean:

$$\hat{\mu} = \left(\frac{n}{\sigma^2}\bar{x} + \frac{1}{\tau^2}\theta\right)\left(\frac{n}{\sigma^2} + \frac{1}{\tau^2}\right)^{-1}$$

Posterior variance:

$$\hat{\sigma}^2 = \left(\frac{n}{\sigma^2} + \frac{1}{\tau^2}\right)^{-1}$$

Credible Interval for the Normal Posterior

The credible set is defined as

$$\{\theta \mid \exp\left(-(\mu - \hat{\mu})^2/2\hat{\sigma}^2\right) > k\}$$

Since the posterior distribution is normal,

$(1 - \alpha)$ level credible interval

$$(\hat{\mu} - z_{\alpha/2}\hat{\sigma}, \hat{\mu} + z_{\alpha/2}\hat{\sigma})$$

Normal Quantiles

$$z_{\alpha/2}$$

denotes the $(1-\alpha/2)$-th quantile of the standard normal distribution:

$$N(0,1)$$

Exercise

Repeat the above calculation using the non-informative prior

$$\pi(\theta) = c$$

Example: Binomial Probability

Suppose

$$X \sim Bin(n, p)$$

Prior:

$$p \sim Beta(\alpha, \beta)$$

Posterior distribution:

$$\pi(p \mid x) \sim Beta(nx + \alpha, n(1 - x) + \beta)$$

HPD Interval for the Binomial Case

The Beta posterior distribution is generally not symmetric around its mean.

Symmetry occurs only if

$$\alpha = \beta$$

Therefore,

- HPD intervals cannot usually be obtained analytically
- Numerical methods are typically used to compute HPD intervals

Conjugate Priors

In many Bayesian models, priors are chosen so that

posterior distribution belongs to the same family

Such priors are called

conjugate priors

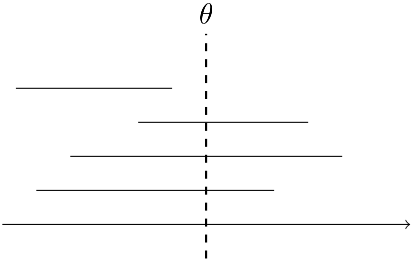
Why Conjugate Priors Are Useful

- Posterior calculations become much simpler
- In sequential data analysis, the posterior after observing data can be used as the prior for the next observation
- The prior can be interpreted as representing information from previously observed data

Frequentist Confidence Interval: Visual Interpretation

Key Idea: Parameter is fixed, interval is random

- True parameter θ is fixed but unknown
- Different samples produce different intervals
- Some intervals contain θ, some do not

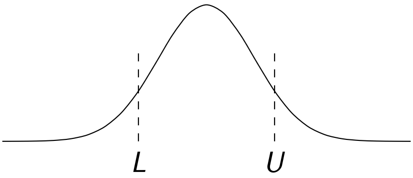


In repeated sampling, about $(1-\alpha)$ proportion of intervals contain θ

Bayesian Credible Interval: Visual Interpretation

Key Idea: Data is fixed, parameter is random

- Observed data is fixed
- Parameter θ has a posterior distribution
- Interval contains $(1 - \alpha)$ probability mass



$$P(\theta \in (L, U) \mid x) = 1 - \alpha$$

Standard Bootstrap as Multinomial Weights

- In standard bootstrap, we resample n observations with replacement.
- Let k_i be the number of times X_i is selected.
- Each draw selects X_i with probability $\frac{1}{n}$.
- Hence:

$$k_i \sim \text{Binomial}\left(n, \frac{1}{n}\right)$$

- Since $\sum_{i=1}^{n} k_i = n$, the vector:

$$\mathbf{k} = (k_1, \ldots, k_n) \sim \text{Multinomial}\left(n; \frac{1}{n}, \ldots, \frac{1}{n}\right)$$

Moments of Bootstrap Weights

- For multinomial weights:

$$E[k_i] = 1$$

$$\mathrm{Var}(k_i) = 1 - \frac{1}{n}$$

$$\mathrm{Cov}(k_i, k_j) = -\frac{1}{n}, \quad i \neq j$$

Bootstrap Estimator (Mean)

- Let parameter:

$$\theta = \mu = \int x \, dF$$

- Bootstrap estimator:

$$\hat{\theta}^* = \frac{1}{n} \sum_{i=1}^{n} k_i X_i$$

- Expectation:

$$E(\hat{\theta}^*) = \bar{X}$$

- Using multinomial properties:

$$n^2 \mathrm{Var}(\hat{\theta}^*) = \sum X_i^2 \mathrm{Var}(k_i) + 2 \sum_{i<j} X_i X_j \mathrm{Cov}(k_i, k_j)$$

- This simplifies to:

$$\mathrm{Var}(\hat{\theta}^*) = \frac{1}{n} \sum (X_i - \bar{X})^2$$

Bootstrap for General Statistics

- Empirical distribution function:

$$\hat{F}(x) = \frac{1}{n}\sum_{i=1}^{n} k_i \mathbf{1}(X_i \leq x)$$

- This is the bootstrap version of the empirical CDF.
- For any statistic $T(F)$:

$$T(\hat{F}) = T^*$$

- Example:

$$T(F) = \int x\, dF \quad \Rightarrow \quad T(\hat{F}) = \bar{X}^*$$

Motivation for Bayesian Bootstrap

- Standard bootstrap assigns **multinomial (discrete) weights**.
- These weights introduce discreteness in resampling.
- Idea:
  - Replace discrete weights with **continuous weights**
  - This leads to smoother variability
- This motivates the **Bayesian Bootstrap**
- Beta distribution can be thought as a continuous extension for binomial proportions. And Dirichlet distribution further generalizes this concept to multiple variables. So we generally use this distribution for Bayesian Bootstrap.

Dirichlet Distribution

Definition

The Dirichlet distribution of order $n \geq 2$ with parameters $\alpha_1, \ldots, \alpha_n > 0$ has a probability density function:

$$f(p_1, \ldots, p_n; \alpha_1, \ldots, \alpha_n) = \frac{1}{\mathrm{B}(\alpha)} \prod_{i=1}^{n} p_i^{\alpha_i - 1}$$

The Normalizing Constant

The multivariate Beta function $\mathrm{B}(\alpha)$ is defined in terms of Gamma functions:

$$\mathrm{B}(\alpha) = \frac{\prod_{i=1}^{n} \Gamma(\alpha_i)}{\Gamma\left(\sum_{i=1}^{n} \alpha_i\right)}$$

Dirichlet Distribution

- The density is defined on:

$$p_i > 0, \quad \sum_{i=1}^{n} p_i = 1$$

- Density:

$$f(p_1, \ldots, p_n) = \frac{\Gamma\left(\sum_{i=1}^{n} \alpha_i\right)}{\prod_{i=1}^{n} \Gamma(\alpha_i)} \prod_{i=1}^{n} p_i^{\alpha_i - 1}$$

- $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$
- $E[p_i] = \dfrac{\alpha_i}{\sum_{j=1}^{n} \alpha_j}$
- Symmetric case: $\alpha_1 = \cdots = \alpha_n = 1$ gives uniform distribution over simplex
- If $e_i \sim \text{Exp}(1)$ then $\dfrac{e_i}{\sum e_i} \sim \text{Dirichlet}(1, \ldots, 1)$

Moments of Dirichlet Distribution

- Let $\alpha_0 = \sum_{i=1}^{n} \alpha_i$
- Second moment:

$$E[p_i^2] = \frac{\alpha_i(\alpha_i + 1)}{\alpha_0(\alpha_0 + 1)}$$

- Variance:

$$\mathrm{Var}(p_i) = \frac{\alpha_i(\alpha_0 - \alpha_i)}{\alpha_0^2(\alpha_0 + 1)}$$

- Covariance $(i \neq j)$:

$$\mathrm{Cov}(p_i, p_j) = -\frac{\alpha_i \alpha_j}{\alpha_0^2(\alpha_0 + 1)}$$

- Key insight:
  - Components are **negatively correlated**
  - Due to constraint $\sum p_i = 1$

Interpretation of Dirichlet Weights

- Bayesian bootstrap assigns **random weights** to observations
- Unlike multinomial weights:
  - These are **continuous**
  - Avoid repeated sample configurations
- Key idea:
  - We are placing a **random distribution** on observed data
- This produces **smoother variability** in statistics

Incorporating Prior Beliefs

- Dirichlet parameters α_i allow prior information
- If some observations are more reliable:
  - Assign larger α_i
- Effect:
  - Higher weight assigned to important observations
- Classical Bayesian bootstrap:

$$\alpha_1 = \cdots = \alpha_n = 1$$

  - Leads to non-informative weights

Example: Sample Mean

- Consider estimator:

$$\hat{\theta}(p) = \sum_{i=1}^{n} p_i x_i$$

- This is a **weighted mean**
- Using Dirichlet weights:

$$(p_1, \ldots, p_n) \sim \text{Dirichlet}(1, \ldots, 1)$$

Expectation of the Estimator

- Using:

$$E(p_i) = \frac{1}{n}$$

- Then:

$$E[\hat{\theta}(p)] = \sum x_i E(p_i) = \frac{1}{n} \sum x_i = \bar{X}$$

- So:
  - Bayesian bootstrap mean is **unbiased**

Variance of the Estimator

- Use:

$$\mathrm{Var}(p_i) = \frac{n-1}{n^2(n+1)}, \quad \mathrm{Cov}(p_i, p_j) = -\frac{1}{n^2(n+1)}$$

- Then:

$$\mathrm{Var}(\hat{\theta}(p)) = \sum x_i^2 \mathrm{Var}(p_i) + 2\sum_{i<j} x_i x_j \mathrm{Cov}(p_i, p_j)$$

- After simplification:

$$\mathrm{Var}(\hat{\theta}(p)) = \frac{1}{n(n+1)} \sum (x_i - \bar{X})^2$$

- Compare with classical bootstrap:
  - Very similar structure
- Insight:
  - Bayesian bootstrap behaves almost like standard bootstrap

Example: Sample Variance

- Consider the statistic:

$$\hat{\sigma}^2(p) = \sum_{i=1}^{n} p_i x_i^2 - \left(\sum_{i=1}^{n} p_i x_i \right)^2$$

- This is the **weighted variance functional**
- Using Dirichlet weights:

$$(p_1, \ldots, p_n) \sim \text{Dirichlet}(1, \ldots, 1)$$

- Goal:
  - Study $E[\hat{\sigma}^2(p)]$

Expectation of Sample Variance

- Using:

$$E(p_i) = \frac{1}{n}, \quad E(p_i^2) = \frac{2}{n(n+1)}, \quad E(p_i p_j) = \frac{1}{n(n+1)}$$

- After simplification:

$$E[\hat{\sigma}^2(p)] = \frac{1}{n+1} \sum_{i=1}^{n} (x_i - \bar{X})^2$$

- Interpretation:
  - Slightly **biased downward**
  - Shrinkage factor:

$$\frac{n}{n+1}$$

- Insight:
  - Acts like a **regularized variance estimator**

Dirichlet Distribution and Bayesian Bootstrap

Connection to Bayesian Bootstrap

- Instead of resampling data points (frequentist bootstrap), assign random weights to observed data
- Let observed data be $x_1, \ldots, x_n$
- Assign weights:

$$(w_1, \ldots, w_n) \sim \text{Dirichlet}(1, 1, \ldots, 1)$$

- Then compute statistics using weighted sample:

$$\hat{\theta} = \sum_{i=1}^{n} w_i x_i$$

Question What if we are trying to bootstrap for some statistic other than the sample mean?

The Plug-in Approach

The Statistical Functional

- Most statistics θ can be seen as a **functional** T of a distribution F.
- Example: Mean $= \int x dF(x)$, Median $= F^{-1}(0.5)$.

The Random ECDF

- Each Dirichlet draw $\mathbf{w}$ defines a random discrete distribution F^*:

$$F^*(x) = \sum_{i=1}^{n} w_i I(x_i \leq x)$$

- The "Plug-in" estimate is simply: $\hat{\theta}^* = T(F^*)$.

Example: Estimating the Median

To find the posterior of the median:

1. Sort your observed data: $x_{(1)} < x_{(2)} < \cdots < x_{(n)}$.
2. Draw weights $(w_1, \ldots, w_n) \sim \text{Dirichlet}(1, \ldots, 1)$.
3. Find the index j where the cumulative weight first hits 0.5:

$$\sum_{i=1}^{j-1} w_{(i)} < 0.5 \quad \text{and} \quad \sum_{i=1}^{j} w_{(i)} \geq 0.5$$

4. The estimate $\hat{\theta}^* = x_{(j)}$.

Note: Continuous weights provide a smoother posterior than the discrete frequentist "jumps."

Numerical Implementation

To draw $(w_1, \ldots, w_n) \sim \text{Dirichlet}(1, \ldots, 1)$ efficiently:

1. Generate n independent draws from an Exponential distribution:
$$u_1, \ldots, u_n \sim \text{Exp}(1)$$
2. Normalize them to sum to 1:
$$w_i = \frac{u_i}{\sum_{j=1}^{n} u_j}$$

The Loop: Repeat this 10,000 times. Each iteration gives you one $\hat{\theta}^*$. The resulting histogram is your **Bayesian Posterior Distribution**.

Limitations

- The Bayesian Bootstrap is restricted to the support of the observed data:
$$\text{Support}(F^*) = \{x_1, \ldots, x_n\}$$
- **Issue with Sample Maxima:** The posterior for the maximum will always be exactly $\max(x_i)$ with probability 1.
- **Issue with $I(\mu > 0)$:** If all observed $x_i < 0$, then $P(\mu > 0|\text{Data}) = 0$, even if the true population mean is positive.

Solution: Move to a **Dirichlet Process** which allows for unobserved values via a base Distribution G_0. This allows the model to assign probability to unobserved values and provides a mechanism to integrate prior beliefs into the bootstrap distribution.

The Dirichlet Process: A Semi-Parametric Solution

- The DP is a **distribution over distributions**, denoted as $G \sim DP(\alpha, G_0)$.
- It addresses the support limitation of the Bayesian Bootstrap using two parameters:
  - **Base Distribution (G_0):** The "center" of the process. It provides support for **unobserved values**, allowing $P(\mu > 0) > 0$ even if all observed $x_i < 0$.
  - **Concentration Parameter (α):** Controls the "strength" of the prior.

Posterior Consistency

Given data $X = \{x_1, \ldots, x_n\}$, the posterior is also a DP:

$$G|X \sim DP\left(\alpha + n, \frac{\alpha}{\alpha + n} G_0 + \frac{n}{\alpha + n} \hat{F}_n\right)$$

Where $\hat{F}_n$ is the empirical distribution. As $\alpha \to 0$, we recover the Bayesian Bootstrap.

The Stick-Breaking Process: Partitioning Probability

The **Sethuraman (1994)** construction provides a recipe for building a random distribution $G \sim DP(\alpha, G_0)$. First, we partition a "unit stick" of total probability 1:

1. **Successive Proportions:** Draw an infinite sequence of random variables from a Beta distribution:

$$\beta_k \sim \text{Beta}(1, \alpha), \quad k = 1, 2, \ldots$$

2. **Breaking the Stick:** Each weight π_k is a portion of what remains:
   - $\pi_1 = \beta_1$
   - $\pi_2 = \beta_2(1 - \beta_1)$
   - $\pi_k = \beta_k \prod_{i=1}^{k-1}(1 - \beta_i)$

The Role of α (Concentration)

- **Small α:** Large pieces are broken off early; G is concentrated on a few atoms (sparse).
- **Large α:** Small pieces are broken off; mass is spread across many atoms (closer to G_0).

The Stick-Breaking Process: Atoms and Support

Once weights are determined, we assign them to locations (atoms) in the state space:

1. **Sampling Locations:** Draw independent atoms from the base distribution:
$$\theta_k \sim G_0, \quad k = 1, 2, \ldots$$
2. **Constructing the Measure:** The random distribution G is the infinite sum of these weighted point masses:
$$G = \sum_{k=1}^{\infty} \pi_k \delta_{\theta_k}$$

- **Infinite Support:** Because θ_k are drawn from G_0, the support of G is the support of G_0, not just the observed data.
- **Discreteness:** Even if G_0 is continuous, G is **almost surely discrete**.
- **Computation:** In practice, we truncate the sum at a finite K such that $\sum_{k=1}^{K} \pi_k \approx 1$.

The DP as a Generalization of the BB

The Bayesian Bootstrap is a **limiting case** of the Dirichlet Process. We can bridge them by considering the posterior distribution:

- **The DP Posterior:** Given data $x_1, \ldots, x_n$, the posterior is:

$$G|\text{Data} \sim DP\left(\alpha + n, \frac{\alpha}{\alpha + n} G_0 + \frac{n}{\alpha + n} \hat{F}_n\right)$$

- **The BB Limit:** As the concentration parameter $\alpha \to 0$:

$$G|\text{Data} \to DP(n, \hat{F}_n)$$

Mechanism of the Bayesian Bootstrap

In the limit $\alpha \to 0$, the base distribution G_0 is ignored. The atoms θ_k are no longer drawn from a prior; they are **fixed** to the observed data points $\{x_1, \ldots, x_n\}$.

Implementation: The DP-Posterior Mixture

To solve the support limitation, we represent the posterior as a mixture of the observed data and the prior distribution G_0:

1. **Partition the Mass:** Allocate total probability 1 between the data and the prior:

$$W_{data} = \frac{n}{\alpha + n}, \quad W_{prior} = \frac{\alpha}{\alpha + n}$$

2. **The Predictive Distribution:** A new "Bootstrap" draw x^* follows:

$$x^* \sim \underbrace{\frac{n}{\alpha + n}\hat{F}_n}_{\text{Observed Data}} + \underbrace{\frac{\alpha}{\alpha + n}G_0}_{\text{Prior Distribution}}$$

3. **Functionals of Interest:** To estimate $\mu = E[G]$, we sample:

$$\mu^* = (1 - w_{new})\left(\sum_{i=1}^{n} \tilde{w}_i x_i\right) + w_{new}\bar{\theta}$$

where $\bar{\theta}$ is a mean draw from G_0 and $w_{new} \sim \text{Beta}(\alpha, n)$.

Calculating the Estimate: The DP-Bootstrap Algorithm

To estimate a functional $\theta = f(G)$ (e.g., the mean or variance) using the Dirichlet Process, we follow a simulation-based approach:

1. **Sample the Mixture Weight:** Draw the relative "prior weight" from a Beta distribution:
$$w_0 \sim \text{Beta}(\alpha, n)$$
(This represents the proportion of the posterior assigned to the prior G_0.)
2. **Sample Data Weights:** Draw n weights for the observed data:
$$(w_1, \ldots, w_n) \sim (1 - w_0) \times \text{Dirichlet}(1, \ldots, 1)$$
3. **Generate the Prior Component:** Since G_0 is a distribution, we represent its contribution by its own mean (or a sample):
$$\theta_{prior} \sim G_0 \quad \text{(or use the expected value } E[G_0])$$

Calculating the Estimate: The DP-Bootstrap Algorithm

The Composite Estimate

The simulated estimate for a single "bootstrap" iteration is:

$$\mu^* = \underbrace{\sum_{i=1}^{n} w_i x_i}_{\text{Data Contribution}} + \underbrace{w_0 \theta_{prior}}_{\text{Prior Contribution}}$$

The Plug-in Estimate

The bootstrap estimate for general plugin statistics $T(F)$ is $T(\cdot)$ applied to the combined weighted sample of observed data and prior draws:

$$\theta^* = T\left(\sum_{i=1}^{n} w_i \delta_{x_i} + \frac{w_0}{M} \sum_{j=1}^{M} \delta_{\theta_j}\right)$$

Comparison: Support of the Estimate

Consider the case where we test if $\mu > 0$ given observations $X = \{-2, -1\}$.

Feature	**Bayesian Bootstrap**	**DP-Bootstrap ($\alpha > 0$)**
Support	Exactly $\{-2, -1\}$	$(-\infty, \infty)$ (if G_0 is Normal)
Max Value	$\max(x_i) = -1$	∞ (determined by G_0)
$P(\mu > 0)$	**Exactly 0**	> 0
Weighting	Only re-shuffles data	Blends data with G_0

Summary: By adding the term $w_0 \theta_{prior}$, we allow the posterior mean to migrate outside the convex hull of the observed data. The parameter α acts as a "smoothing" factor—the larger the α, the more the estimate is "pulled" toward the prior distribution G_0.

Discussion: Flexibility of the Dirichlet Process Framework

The transition from the Bayesian Bootstrap to the Dirichlet Process (DP) is not just a mathematical fix for support; it enables a truly **Bayesian workflow**:

- **Conjugacy and Online Learning:** The Dirichlet Process is conjugate to i.i.d. sampling. This allows for **recursive updating**:
  - Yesterday's posterior $DP(\alpha_{post}, G_{post})$ becomes today's prior.
  - This is computationally efficient for streaming data, as we only need to update the concentration parameter $(\alpha + n)$ and the weighted base distribution.
- **Integrating Field Knowledge:** Unlike the "non-informative" Bayesian Bootstrap, the DP allows the researcher to inject **domain expertise**:
  - If the data represents physical constants or known biological limits, G_0 can be tuned to reflect those boundaries.
  - The concentration parameter α acts as a "prior sample size," quantifying exactly how much confidence we have in our expert knowledge relative to the n observed data points.

What is a Markov Chain?

Imagine a particle moving between different "states" (e.g., Weather: Sunny, Cloudy, Rainy). A **Markov Chain** is a mathematical system that hops from one state to another according to specific probabilities.

The "Memoryless" Property (Markov Property)

The future depends **only** on the present, not the past.

$$P(X_{t+1}|X_t, X_{t-1}, \ldots, X_0) = P(X_{t+1}|X_t)$$

To the chain, it doesn't matter *how* it got to the current state; it only matters *where* it is right now.

- **States (S):** The possible values the variable can take.
- **Transitions:** The probability of moving from state i to state j.

How Chains Move: The Transition Matrix

We can organize all possible moves into a **Transition Matrix** (P), where the entry P_{ij} is the probability of moving from state i to state j.

- **The Random Walk:** If we let the chain run for a very long time, where does it end up?
- **Stationary Distribution (π):** For many chains, the proportion of time spent in each state eventually stabilizes.

The Intuition for MCMC

In Bayesian stats, we have a "Target Distribution" (our Posterior). **The Goal of MCMC:** We purposely design a Markov Chain so that its "long-run" stable state (π) is exactly the distribution we want to sample from.

How Chains Move: The Transition Matrix

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Example: A Simple Random Walk

Consider a particle moving on a number line.

- At each step:
  - Move **right** with probability 0.5
  - Move **left** with probability 0.5

Why is this a Markov Chain?

The next position depends only on the current position:

$$P(X_{t+1} = x + 1 \mid X_t = x)$$

It does not depend on how we reached x.

Key Idea: The "state" is simply the current position.

Random Walk as a Markov Chain

Let the state space be:

$$S = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

- From any state i:
  - Move to $i+1$ with probability 0.5
  - Move to $i-1$ with probability 0.5

Transition Structure

$$P_{i,i+1} = 0.5, \quad P_{i,i-1} = 0.5$$

Observation: The rules are identical at every state.

Finite Random Walk: Transition Matrix

Now consider a finite state space:

$$S = \{1, 2, 3\}$$

- From state 2:
  - Go to 1 with probability 0.5
  - Go to 3 with probability 0.5
- From boundaries:
  - $1 \to 2$ with probability 1
  - $3 \to 2$ with probability 1

Transition Matrix

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0.5 & 0 & 0.5 \\ 0 & 1 & 0 \end{bmatrix}$$

Visualizing the Random Walk

$$1 \leftrightarrow 2 \leftrightarrow 3$$

- The particle "bounces" between states
- Over time, it keeps moving indefinitely

Important Insight

Even though the movement is random, the long-run behavior is structured.

This is exactly what MCMC exploits.

Why This Matters for MCMC

- A random walk is a simple Markov Chain
- It generates a sequence of dependent samples

Key Insight

If we design the transition probabilities carefully, the chain will:

- Visit important regions more often
- Approximate a target distribution

MCMC = Smart Random Walk

Transition Probability Matrix

For a Markov chain with state space $S = \{1, 2, \ldots, m\}$:

Definition

The transition matrix P is defined by:

$$P_{ij} = P(X_{t+1} = j \mid X_t = i)$$

- Each row sums to 1:

$$\sum_j P_{ij} = 1$$

- All entries are non-negative

Interpretation: Row i gives probabilities of moving from state i.

A Smart Markov Chain for Binomial

Suppose we want a chain whose long-run behavior is:

$$\pi(k) \propto \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n$$

- Example: $n = 10$, $p = 0.5$
- Goal: Spend more time near likely values (around np)

Step 1: Only Local Moves

From state k, the chain can:

- Move to $k+1$ (increase)
- Move to $k-1$ (decrease)
- Stay at k

Interpretation

Think of k as "number of successes".

- Increase $k \to$ gain a success
- Decrease $k \to$ lose a success

Step 2: Bias the Movement

We choose:

$$P(k \to k+1) = \frac{(n-k)p}{n}$$

$$P(k \to k-1) = \frac{k(1-p)}{n}$$

$$P(k \to k) = \frac{k}{n}p + \frac{n-k}{n}(1-p)$$

Observe the probabilities are chosen in such a way so that it tries to move towards the middle.

Why This Works

- If k is too small:
  - $(n-k)p$ is large $\to$ pushes upward
- If k is too large:
  - $k(1-p)$ is large $\to$ pushes downward

Key Insight

The chain automatically drifts toward np (the Binomial mean).

Long-Run Behavior

- The chain keeps moving forever
- But visits states in a stable proportion

Result

The stationary distribution of this chain is exactly:

$$\pi(k) = \text{Binomial}(n, p)$$

What is a Stationary Distribution?

Let π be a probability distribution over the state space.

Definition

π is called a **stationary distribution** if:

$$\pi P = \pi$$

- Applying the transition does not change the distribution
- The distribution remains *unchanged over time* even if the markov cain keeps running.

Stationarity via Total Probability

Let $\pi(j) = P(X_t = j)$.

$$\begin{aligned} P(X_{t+1} = j) &= \sum_i P(X_{t+1} = j, X_t = i) \\ &= \sum_i P(X_{t+1} = j \mid X_t = i)\, P(X_t = i) \\ &= \sum_i \pi(i)\, P_{ij} \end{aligned}$$

Stationarity Condition

If the distribution does not change over time:

$$\pi(j) = \sum_i \pi(i)\, P_{ij}$$

Intuition

- Imagine many particles evolving according to the chain
- If their distribution is π:
  - After one step $\rightarrow$ still π
  - After many steps $\rightarrow$ still π

Key Idea

Stationary $=$ **equilibrium** of the Markov chain

Connection to Our Construction

Recall:

$$P_{ij} = \pi(j)$$

Then:

$$(\pi P)_j = \sum_i \pi(i) P_{ij} = \sum_i \pi(i)\pi(j) = \pi(j)$$

Conclusion

The chain was constructed so that π is stationary.

Why Do We Care?

- If a chain has stationary distribution π
- And we run it for a long time

Big Idea

The samples we generate will behave like draws from π

This is the foundation of MCMC.

Checking Stationarity

We claim:

$$\pi(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Definition

π is stationary if $(\pi P)(k) = \pi(k)$

$$\begin{aligned}(\pi P)(k) = \pi(k-1)\frac{(n-k+1)p}{n} \\ + \pi(k)\left(\frac{k}{n}p + \frac{n-k}{n}(1-p)\right) \\ + \pi(k+1)\frac{(k+1)(1-p)}{n}\end{aligned}$$

Simplification

Using:

$$\pi(k-1)(n-k+1)p = k\,\pi(k)$$
$$\pi(k+1)(k+1)(1-p) = (n-k)\,\pi(k)$$

$$\begin{aligned}(\pi P)(k) &= \frac{1}{n}\Big[k\pi(k) + \pi(k)(kp + (n-k)(1-p)) + (n-k)\pi(k)\Big] \\ &= \frac{1}{n}\cdot n\pi(k) \\ &= \pi(k)\end{aligned}$$

Conclusion

Binomial(n, p) is a stationary distribution.

Is the Stationary Distribution Unique?

Important Note

Showing $(\pi P)(k) = \pi(k)$ only proves that π is **a** stationary distribution.

- It does **not** guarantee:
  - Uniqueness
  - Convergence to π from any starting point

What Do We Need?

Additional conditions on the Markov chain:

- Irreducibility : ensures every state is reachable from everywhere.
- Aperiodicity : ensures the chain will not get stuck in a loop.

Good news: For such chains, the stationary distribution is unique and the chain converges to it.

Periodicity: Does the Chain Have a Rhythm?

A state i is periodic if the chain can only return to it at fixed, predictable intervals.

The Period (d)

The period $d(i)$ of a state i is the greatest common divisor (GCD) of all possible path lengths from i back to i:

$$d(i) = \gcd\{n > 0 : P_{ii}^{(n)} > 0\}$$

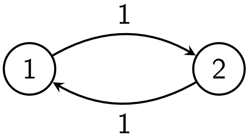
- **Periodic:** If $d(i) > 1$. The chain is "stuck" in a cycle.
- **Aperiodic:** If $d(i) = 1$. The return times are irregular.

Fact: In an irreducible chain, all states share the same period. If one state is aperiodic, the whole chain is aperiodic.

Example 1: The Periodic "Tennis Match"

Consider a 2-state chain $S = \{1, 2\}$ where the particle *must* move to the other state:

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$



- To return to State 1, the paths must be: $1 \to 2 \to 1$ (2 steps), $1 \to 2 \to 1 \to 2 \to 1$ (4 steps), etc.
- Possible return times: $\{2, 4, 6, 8, \dots\}$
- $d(1) = \text{gcd}(2, 4, 6, \dots) = 2$.

Why it fails to converge: The chain never "settles." It oscillates forever between states based on whether the step is even or odd.

Example 2: How to Break the Rhythm

Aperiodicity is required for a chain to converge to a stationary distribution.
The easiest way to break a cycle is a **self-loop**.
Modify the previous chain so there is a small chance to stay in State 1:

$$P = \begin{bmatrix} 0.1 & 0.9 \\ 1 & 0 \end{bmatrix}$$

- Now, the particle can return to State 1 in:
  - **1 step:** $1 \to 1$ (staying put)
  - **2 steps:** $1 \to 2 \to 1$
- $d(1) = \gcd(1, 2) = 1$.

The "Self-Loop" Rule

If any state in an irreducible chain has $P_{ii} > 0$, the entire chain is **aperiodic**.

Example: The Restricted Random Walk

Consider a particle moving on a finite state space $S = \{1, 2, 3\}$. At each step, the particle moves according to these rules:

- From State 1: 60% chance to stay, 40% chance to move to 2.
- From State 2: 60% chance to move to 1, 40% chance to move to 3.
- From State 3: 60% chance to move to 2, 40% chance to stay.

Questions

1. What is the 3×3 transition matrix P.
2. Write down the system of equations for the stationary distribution $\pi = [\pi_1, \pi_2, \pi_3]$ using the condition:

$$\pi P = \pi$$

3. Solve for π (Don't forget the normalization $\sum \pi_i = 1$).

Intuition: The "pull" toward State 1 (0.6) is stronger than the "push" toward State 3 (0.4). Which state should have the highest probability?

The Waiting Time Problem

Suppose we flip a fair coin repeatedly. How many flips, on average, does it take to see a specific pattern?

- Consider the pattern: **H H** (Heads followed by Heads).
- Does it take the same amount of time as **H T**?

The Markov Chain Approach

We define states based on our current progress toward the pattern:

- **State 0:** Start (no "H" recently seen).
- **State 1:** One "H" seen.
- **State 2:** "H H" achieved (*Absorbing State*).

Transition Matrix for Pattern "HH"

Let $p = 0.5$ (Heads) and $q = 0.5$ (Tails).

- From State 0: $H \to$ State 1, $T \to$ State 0.
- From State 1: $H \to$ State 2 (Goal!), $T \to$ State 0 (Reset).

Transition Matrix P

$$P = \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0.5 & 0 & 0.5 \\ 0 & 0 & 1 \end{bmatrix}$$

Note: If you are at State 1 and get a Tails, you lose all progress and go back to State 0.

Solving for Expected Time (HH)

Let E_i be the expected number of *additional* flips needed to reach State 2, starting from State i.

Using **First-Step Analysis**:

1. $E_0 = 1 + 0.5(E_0) + 0.5(E_1)$
2. $E_1 = 1 + 0.5(E_0) + 0.5(E_2)$
3. $E_2 = 0$ (We are already there)

The Solution

Solving the system of equations:

$$E_1 = 1 + 0.5E_0$$

Substitute into (1):

$$E_0 = 1 + 0.5E_0 + 0.5(1 + 0.5E_0)$$

$$E_0 = 1.5 + 0.75E_0 \implies 0.25E_0 = 1.5 \implies \mathbf{E_0 = 6}$$

The "HT" Paradox

If we perform the same analysis for **H T**:

- From State 1 (having an 'H'):
  - $T \rightarrow$ State 2 (Success)
  - $H \rightarrow$ State 1 (You still have an 'H', so you don't reset!)

Comparison

- Expected time for **HH**: **6 flips**
- Expected time for **HT**: **4 flips**

Theoretical Insight: Patterns that can "overlap" with themselves (like HH) take longer to appear because failures can reset progress more severely.

Example: The Gambler's Ruin

Imagine a gambler starts with \$$k$ and plays a game against an opponent (or the "house").

- In each round:
  - Wins \$1 with probability p
  - Loses \$1 with probability $q = 1 - p$
- The game ends when the gambler reaches:
  - \$$N$ (The Target Goal)
  - \$0 (The "Ruin")

Why is this a Markov Chain?

The gambler's wealth at time $t + 1$ depends only on their wealth at time t. States 0 and N are **absorbing states**—once you enter them, you never leave.

Transition Structure of Gambler's Ruin

Let the state space be $S = \{0, 1, 2, \ldots, N\}$.

- For $0 < i < N$:

$$P_{i,i+1} = p, \quad P_{i,i-1} = q$$

- At the boundaries:

$$P_{0,0} = 1, \quad P_{N,N} = 1$$

The Transition Matrix P

$$P = \begin{bmatrix} 1 & 0 & 0 & \ldots & 0 \\ q & 0 & p & \ldots & 0 \\ 0 & q & 0 & p & \ldots \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \ldots & 0 & 1 \end{bmatrix}$$

Long-Run Outcome: Probability of Ruin

In this specific chain, the "long-run" doesn't lead to a single stationary distribution π that visits all states. Instead, the particle eventually gets "trapped" at 0 or N.

- Let P_i be the probability of reaching N starting from i.
- Using the law of total probability (one-step analysis):

$$P_i = pP_{i+1} + qP_{i-1}$$

The Solution (for $p \neq 0.5$)

The probability of reaching the goal N is:

$$P_i = \frac{1 - (q/p)^i}{1 - (q/p)^N}$$

MCMC Connection: This shows how specific transition rules (p vs q) drastically change where the chain "ends up."

Two-Step Transition Probabilities

Definition

$$P_{ij}^{(2)} = P(X_{t+2} = j \mid X_t = i)$$

Using law of total probability over intermediate state k:

$$\begin{aligned} P_{ij}^{(2)} &= \sum_k P(X_{t+2} = j, X_{t+1} = k \mid X_t = i) \\ &= \sum_k P(X_{t+2} = j \mid X_{t+1} = k)\, P(X_{t+1} = k \mid X_t = i) \\ &= \sum_k P_{kj}\, P_{ik} \end{aligned}$$

Matrix Representation

Recall matrix multiplication:

$$(P^2)_{ij} = \sum_k P_{ik} P_{kj}$$

Comparing with:

$$P^{(2)}_{ij} = \sum_k P_{ik} P_{kj}$$

Conclusion

$$P^{(2)} = P^2$$

Interpretation: Two-step transitions are obtained by composing one-step transitions.

n-step Transition Probabilities

Claim

$$P^{(n)} = P^n$$

Proof by induction:

- **Base case ($n = 1$):**
$$P^{(1)} = P$$
- **Base case ($n = 2$):**
$$P^{(2)} = P^2 \quad \text{(shown previously)}$$
- **Inductive step:** Assume $P^{(n)} = P^n$
- Then:
$$P_{ij}^{(n+1)} = \sum_k P_{ik}^{(n)} P_{kj}$$
- Hence:
$$P^{(n+1)} = P^{(n)} P = P^n P = P^{n+1}$$

How Do We Combine Steps?

Question

Suppose you know:

- $P_{ik}^{(m)}$: probability of going from i to k in m steps
- $P_{kj}^{(n)}$: probability of going from k to j in n steps for all k

How do you compute:

$$P_{ij}^{(m+n)} = P(X_{t+m+n} = j \mid X_t = i)\,?$$

How Do We Combine Steps?

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How do you compute:

$$P_{ij}^{(m+n)} = P(X_{t+m+n} = j \mid X_t = i)\,?$$

Idea: Where could we be after m steps?

- After m steps, the chain must be in some state k
- Then it moves from k to j in n steps

Combining the Steps

Using law of total probability over intermediate state k:

$$
\begin{aligned}
P_{ij}^{(m+n)} &= \sum_k P(X_{t+m+n} = j, X_{t+m} = k \mid X_t = i) \\
&= \sum_k P(X_{t+m+n} = j \mid X_{t+m} = k)\, P(X_{t+m} = k \mid X_t = i) \\
&= \sum_k P_{kj}^{(n)}\, P_{ik}^{(m)}
\end{aligned}
$$

Chapman–Kolmogorov Equation

$$P_{ij}^{(m+n)} = \sum_k P_{ik}^{(m)} P_{kj}^{(n)}$$

Interpretation: Break the journey at time $t + m$.

What Happens in the Long Run?

Question

What happens to the n-step transition probabilities:

$$P_{ij}^{(n)}$$

as $n \to \infty$?

- Does it depend on the starting state i?
- Does it converge to some fixed value?

Goal: Understand the long-term behavior of the chain.

Understanding P^n via Eigenvalues

Let P be the transition matrix.

Key Idea

The behavior of P^n is governed by the eigenvalues of P.

If λ is an eigenvalue of P, then:

$$\lambda^n \text{ appears in } P^n$$

- Eigenvalues satisfy $|\lambda| \leq 1$
- If $|\lambda| < 1$, then $\lambda^n \to 0$
- Only $\lambda = 1$ survives as $n \to \infty$

Conclusion

Long-term behavior of P^n is determined by eigenvalue 1.

Why Are Eigenvalues ≤ 1?

Let P be a transition matrix and suppose:

$$Px = \lambda x$$

Choose k such that:

$$|x_k| = \max_i |x_i|$$

Then:

$$\begin{aligned}
|\lambda||x_k| &= \left|\sum_j P_{kj} x_j\right| \\
&\leq \sum_j P_{kj} |x_j| \\
&\leq \sum_j P_{kj} |x_k| \\
&= |x_k|
\end{aligned}$$

Long-Run Behavior

Under suitable conditions (irreducible, aperiodic) a chain has only one eigenvector of probabilities corresponding to the eigenvalue 1 all the remaining eigenvalues converge to 0. Hence :

$$P_{ij}^{(n)} \longrightarrow \pi(j) \quad \text{as } n \to \infty$$

- The limit does **not** depend on the starting state i
- Every row of P^n becomes identical

Interpretation

The chain "forgets" where it started and converges to the stationary distribution.

$$\Rightarrow \quad P^n \to \left[\pi, \pi, \cdots, \pi\right]^T$$

Designing Markov Chains

So far:

- If a chain has stationary distribution π
- Then $P^n \to \pi$

New Question

How do we **construct** a Markov chain with a desired stationary distribution π?

Designing Markov Chains

So far:

- If a chain has stationary distribution π
- Then $P^n \to \pi$

New Question

How do we **construct** a Markov chain with a desired stationary distribution π?

Idea: Enforce a stronger condition than stationarity.

Reversibility (Detailed Balance)

Definition

A Markov chain is **reversible** w.r.t. π if:

$$\pi(i)P_{ij} = \pi(j)P_{ji}$$

- Probability flow from $i \to j$
- Equals flow from $j \to i$

Interpretation: No net flow between states at equilibrium.

Why Reversibility Works

Start from:

$$\sum_i \pi(i) P_{ij}$$

Using detailed balance:

$$\begin{aligned}\sum_i \pi(i) P_{ij} &= \sum_i \pi(j) P_{ji} \\ &= \pi(j) \sum_i P_{ji} \\ &= \pi(j)\end{aligned}$$

Conclusion

$$\pi P = \pi$$

So reversibility $\Rightarrow$ stationarity

Why Do We Use Reversibility?

- Directly solving $\pi P = \pi$ is hard
- Reversibility is easier to enforce

Key Idea

Instead of solving for π, we design P so that:

$$\pi(i)P_{ij} = \pi(j)P_{ji}$$

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Key Idea

Instead of solving for π, we design P so that:

$$\pi(i)P_{ij} = \pi(j)P_{ji}$$

This is the foundation of:

- Metropolis–Hastings
- Gibbs Sampling

Sampling from a Target Distribution

Goal

We want to sample from a target distribution:

$$\pi(x)$$

- Often known only up to a constant:

$$\pi(x) \propto f(x)$$

- Direct sampling may be difficult

Idea

Use a simpler distribution that we can sample from.

Rejection Sampling Setup

Choose:

- Proposal distribution $g(x)$ (easy to sample)
- Constant M such that:

$$f(x) \leq Mg(x) \quad \forall x$$

Interpretation

$Mg(x)$ is an envelope over $f(x)$.

Rejection Sampling Algorithm

Repeat:

- Sample $X \sim g(x)$
- Sample $U \sim \text{Uniform}(0, 1)$
- Accept X if:

$$U \leq \frac{f(X)}{Mg(X)}$$

Otherwise

Reject and repeat.

Unconditional Acceptance Probability

Let $Y \sim g(x)$ and $U \sim \text{Uniform}(0, 1)$.

$$
\begin{aligned}
P\left(U \leq \frac{f(Y)}{Mg(Y)}\right) &= \mathbb{E}\left[P\left(U \leq \frac{f(Y)}{Mg(Y)} \mid Y\right)\right] \\
&= \mathbb{E}\left[\frac{f(Y)}{Mg(Y)}\right] \\
&= \int \frac{f(y)}{Mg(y)} g(y)\, dy \\
&= \frac{1}{M} \int f(y)\, dy \\
&= \frac{1}{M}
\end{aligned}
$$

Distribution of Accepted Samples

Let:

- $Y \sim g(x)$
- $U \sim \text{Uniform}(0, 1)$
- Accept if $U \leq \frac{f(Y)}{Mg(Y)}$

We compute the density of Y given acceptance.

Step 1: Joint probability

$$
\begin{aligned}
P(Y \in A, \text{accept}) &= \int_A g(y)\, P\left(U \leq \frac{f(y)}{Mg(y)} \mid Y = y\right) dy \\
&= \int_A g(y)\, \frac{f(y)}{Mg(y)}\, dy \\
&= \frac{1}{M} \int_A f(y)\, dy
\end{aligned}
$$

Conditional Density of Accepted Samples

Acceptance probability:

$$P(\text{accept}) = \frac{1}{M}$$

So:

$$\begin{aligned} P(Y \in A \mid \text{accept}) &= \frac{P(Y \in A, \text{accept})}{P(\text{accept})} \\ &= \frac{\frac{1}{M} \int_A f(y)\, dy}{\frac{1}{M}} \\ &= \int_A f(y)\, dy \end{aligned}$$

Conclusion

The accepted samples have density $f(x)$

Why Choice of M Matters

Key Result

Acceptance probability $= \frac{1}{M}$

- Number of proposals until acceptance $\sim$ Geometric $\left(\frac{1}{M}\right)$.
- Expected number of trials: $\mathbb{E}[\text{trials}] = M$.

Practical Difficulty

- Need $f(x) \leq Mg(x)$ for all x.
- g must cover entire support of f.
- Finding good (M, g) is often difficult.

Conclusion: Poor choice $\Rightarrow$ very inefficient sampling

Limitations of Rejection Sampling

- Requires $f(x) \leq Mg(x)$ for all x
- Finding good $g(x)$ and small M is difficult
- Efficiency drops in high dimensions

Key Issue

Many proposals are rejected $\Rightarrow$ wasteful

Idea

Instead of independent samples, construct a **Markov chain** whose stationary distribution is $\pi(x)$.

Class Problems: Rejection Sampling

- **Problem 1: Uniform Proposal** Find the value of M.
  Target: $f(x) = 2x,\ x \in [0,1]$
  Proposal: $g(x) = 1,\ x \in [0,1]$
- **Problem 2: Acceptance Rule**
  Target: $f(x) = 2x,\ x \in [0,1]$
  Proposal: $g(x) = 1,\ x \in [0,1]$
- **Problem 3: Thin tails**
  Target: $f(x) = e^{-|x|},\ x \geq 0$
  Proposal: $g(x) = \mathcal{N}(0,1)$
- **Problem 4: Support Mismatch**
  Target: $f(x) = \mathcal{N}(0,1)$
  Proposal: $g(x) = \text{Uniform}[-1,1]$
- **Problem 5: High-Dimensional Case**
  Target: $f(x) = \mathcal{N}(0, I_d)$
  Proposal: $g(x) = \text{i.i.d Exponential}(\theta)$

Markov Chain Monte Carlo (MCMC)

Idea

Generate samples using a Markov chain:

$$X_0 \to X_1 \to X_2 \to \cdots$$

- Each X_{t+1} depends only on X_t
- After many steps, distribution converges to $\pi(x)$

Goal

Design transitions so that:

$$\pi(x) \text{ is stationary}$$

Detailed Balance Condition (Reversibility)

A sufficient condition for $\pi(x)$ to be stationary:

$$\pi(x)P(x \to y) = \pi(y)P(y \to x)$$

Interpretation

Flow from $x \to y$ equals flow from $y \to x$

Strategy

Construct transition probabilities satisfying this condition

Continuous-State Markov Chains (Brief Overview)

- State space is continuous: $X_t \in \mathbb{R}^d$
- Cannot use transition *matrix* anymore
- Use a **transition density / kernel**:

$$P(x \to y) = K(y \mid x)$$

- Evolution:

$$X_{t+1} \sim K(\cdot \mid X_t)$$

Key Idea

Instead of probabilities of jumps, we now work with **densities**

Stationary Distribution (Continuous Case)

A distribution $\pi(x)$ is stationary if:

$$\pi(y) = \int \pi(x)\, K(y \mid x)\, dx$$

- Continuous analogue of $\pi P = \pi$
- Distribution remains unchanged after one step

Interpretation

If $X_t \sim \pi$, then $X_{t+1} \sim \pi$

How to Ensure Stationarity?

A sufficient condition: **Detailed Balance**

$$\pi(x)\,K(y \mid x) = \pi(y)\,K(x \mid y)$$

- Stronger than stationarity
- Much easier to enforce in practice

Why this matters

Metropolis–Hastings is designed to satisfy this condition

Why Metropolis–Hastings?

- Rejection sampling can be inefficient
- Difficult to find good envelope $Mg(x)$

Idea

Construct a **Markov chain** whose stationary distribution is $\pi(x)$

- Reuse previous samples
- Avoid complete rejection of proposals
- Avoid finding an envelope

Markov Chain Construction

Generate a sequence:

$$X_0, X_1, X_2, \ldots$$

- X_{t+1} depends only on X_t
- Transition kernel: $P(x \to y)$

Goal

Design transitions such that:

$$\pi(x) \text{ is stationary}$$

Proposal Distribution

At state $X_t = x$:

- Propose a candidate:

$$Y \sim q(y \mid x)$$

- $q(y \mid x)$ is easy to sample from.

Examples

- Gaussian random walk
- Independence proposal

Acceptance Probability

Accept proposed move with probability:

$$\alpha(x, y) = \min\left(1,\ \frac{\pi(y)\, q(x \mid y)}{\pi(x)\, q(y \mid x)}\right)$$

Using unnormalized density

If $\pi(x) \propto f(x)$:

$$\alpha(x, y) = \min\left(1,\ \frac{f(y)\, q(x \mid y)}{f(x)\, q(y \mid x)}\right)$$

Key: Normalizing constant cancels

Metropolis–Hastings Algorithm

Given $X_t = x$:

1. Sample $Y \sim q(y \mid x)$
2. Sample $U \sim \text{Uniform}(0, 1)$
3. If
$$U \leq \alpha(x, y)$$
   set $X_{t+1} = Y$
4. Otherwise, set $X_{t+1} = x$

Transition Kernel

The transition probability is:

$$P(x \to y) = \begin{cases} q(y \mid x)\,\alpha(x,y), & y \neq x \\ 1 - \int q(z \mid x)\,\alpha(x,z)\,dz, & y = x \end{cases}$$

Interpretation

- Move to y with acceptance probability
- Otherwise stay at x

Detailed Balance: Case 1

Case 1:

$$\frac{\pi(y)\, q(x \mid y)}{\pi(x)\, q(y \mid x)} \leq 1$$

Then:

$$\alpha(x, y) = \frac{\pi(y)\, q(x \mid y)}{\pi(x)\, q(y \mid x)}$$

So:

$$\begin{aligned}
\pi(x) P(x \to y) &= \pi(x)\, q(y \mid x)\, \alpha(x, y) \\
&= \pi(x)\, q(y \mid x)\, \frac{\pi(y)\, q(x \mid y)}{\pi(x)\, q(y \mid x)} \\
&= \pi(y)\, q(x \mid y)
\end{aligned}$$

Detailed Balance: Case 2

Case 2:

$$\frac{\pi(y)\,q(x \mid y)}{\pi(x)\,q(y \mid x)} > 1$$

Then:

$$\alpha(x, y) = 1$$

So:

$$\pi(x)P(x \to y) = \pi(x)\,q(y \mid x)$$

For reverse move:

$$\alpha(y, x) = \frac{\pi(x)\,q(y \mid x)}{\pi(y)\,q(x \mid y)}$$

Hence:

$$\begin{aligned}
\pi(y)P(y \to x) &= \pi(y)\,q(x \mid y)\,\alpha(y, x) \\
&= \pi(y)\,q(x \mid y)\,\frac{\pi(x)\,q(y \mid x)}{\pi(y)\,q(x \mid y)} \\
&= \pi(x)\,q(y \mid x)
\end{aligned}$$

Detailed Balance

Metropolis–Hastings satisfies:

$$\pi(x)P(x \to y) = \pi(y)P(y \to x)$$

Why important?

Ensures $\pi(x)$ is a stationary distribution

Key Idea

Acceptance probability corrects asymmetry in q

Intuition

- If $\pi(y) > \pi(x)$:
  - Move is always accepted
- If $\pi(y) < \pi(x)$:
  - Move accepted with probability:

$$\frac{\pi(y)}{\pi(x)} \times \frac{q(x \mid y)}{q(y \mid x)}$$

Insight

Allows exploration but favors high-density regions

Properties of MH

- Produces correlated samples
- Requires burn-in period
- Thinning (keeping every k-th sample) can reduce correlation
- Converges under mild conditions:
  - irreducibility
  - aperiodicity

Advantage

Works even when $\pi(x)$ is known only up to a constant

Gaussian Random Walk Proposal

A common choice:

$$q(y \mid x) = \mathcal{N}(y; x, \sigma^2 I)$$

- Propose:

$$Y = x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$$

- Small random perturbation around current state

Interpretation

Local exploration of the state space

Symmetry of Gaussian Proposal

For Gaussian random walk:

$$q(y \mid x) = q(x \mid y)$$

Reason

$$\mathcal{N}(y; x, \sigma^2 I) = \mathcal{N}(x; y, \sigma^2 I)$$

Consequence

Acceptance probability simplifies:

$$\alpha(x, y) = \min\left(1,\ \frac{\pi(y)}{\pi(x)}\right)$$

Behavior of Gaussian Random Walk

- If σ is small:
  - High acceptance rate.
  - Slow exploration
- If σ is large:
  - Low acceptance rate.
  - Many rejections

Trade-off

Choose σ to balance:

- exploration vs acceptance

Example: Using Unnormalized Density

Suppose:

$$\pi(x) \propto f(x)$$

Then acceptance probability:

$$\alpha(x, y) = \min\left(1,\ \frac{f(y)}{f(x)}\right)$$

Key Advantage

No need to compute normalizing constant

Independence Sampler

Proposal does not depend on current state:

$$q(y \mid x) = q(y)$$

Acceptance probability:

$$\alpha(x, y) = \min\left(1, \frac{\pi(y)\, q(x)}{\pi(x)\, q(y)}\right)$$

Interpretation

Global proposals instead of local moves

Behavior of Independence Sampler

- Works well if $q(y) \approx \pi(y)$
- Poor choice $\Rightarrow$ many rejections

Comparison

- Gaussian random walk: local exploration
- Independence sampler: global exploration

Problem in High Dimensions

- In high dimensions, Metropolis–Hastings can become inefficient
- Proposal $q(y \mid x)$ must move in $\mathbb{R}^d$
- Difficult to design good global proposals

Issue

- Large moves $\Rightarrow$ low acceptance
- Small moves $\Rightarrow$ slow exploration

Consequence

Mixing becomes very slow in high-dimensional spaces

To deal with this we can try to use the conditional distributions in some way.

A Natural Question

Question

Instead of proposing the full vector $Y \sim q(y \mid x)$, what if we update **one component at a time**?

- Decompose:

$$x = (x_1, x_2, \ldots, x_d)$$

- Update only one coordinate while keeping others fixed

Using Conditional Distributions

Key Idea

Use the conditional distribution as the proposal:

$$q(y_i \mid x) = \pi(y_i \mid x_{-i})$$

- x_{-i} denotes all components except x_i
- Only one coordinate is updated at a time

What Happens to Acceptance Probability?

Recall:

$$\alpha(x, y) = \min\left(1, \frac{\pi(y)\, q(x \mid y)}{\pi(x)\, q(y \mid x)}\right)$$

Substitute conditional proposal

$$q(y \mid x) = \pi(y_i \mid x_{-i})$$

Result

All terms cancel $\Rightarrow$

$$\alpha(x, y) = 1$$

Acceptance Probability Calculation

Consider updating only coordinate i:

$$x = (x_i, x_{-i}), \quad y = (y_i, x_{-i})$$

Proposal

$$q(y \mid x) = \pi(y_i \mid x_{-i})$$

Similarly,

$$q(x \mid y) = \pi(x_i \mid x_{-i})$$

Acceptance ratio

$$\frac{\pi(y)\, q(x \mid y)}{\pi(x)\, q(y \mid x)} = \frac{\pi(y_i, x_{-i})\, \pi(x_i \mid x_{-i})}{\pi(x_i, x_{-i})\, \pi(y_i \mid x_{-i})}$$

Cancellation

Use factorization:

$$\pi(x_i, x_{-i}) = \pi(x_i \mid x_{-i})\,\pi(x_{-i})$$

$$\pi(y_i, x_{-i}) = \pi(y_i \mid x_{-i})\,\pi(x_{-i})$$

Substitute into the ratio:

$$\frac{\pi(y_i \mid x_{-i})\,\pi(x_{-i})\,\pi(x_i \mid x_{-i})}{\pi(x_i \mid x_{-i})\,\pi(x_{-i})\,\pi(y_i \mid x_{-i})}$$

Cancellation

All terms cancel $\Rightarrow 1$

$$\alpha(x, y) = 1$$

Gibbs Sampling

Definition

Gibbs sampling updates each coordinate using its conditional distribution

Given $X_t = x$:

$$
\begin{aligned}
x_1^{(t+1)} &\sim \pi\big(x_1 \mid x_2^{(t)}, \ldots, x_d^{(t)}\big) \\
x_2^{(t+1)} &\sim \pi\big(x_2 \mid x_1^{(t+1)}, x_3^{(t)}, \ldots\big) \\
&\vdots \\
x_d^{(t+1)} &\sim \pi\big(x_d \mid x_1^{(t+1)}, \ldots, x_{d-1}^{(t+1)}\big)
\end{aligned}
$$

Key Property of Gibbs Sampling

Important Result

Every proposed move is accepted

- No rejection step
- No need to compute acceptance ratio

Interpretation

We sample directly from the correct conditional distributions

Transition Kernel (Gibbs)

The transition kernel updates one coordinate at a time:

$$K(y \mid x) = \pi(y_i \mid x_{-i}) \cdot \mathbf{1}(y_{-i} = x_{-i})$$

Interpretation

- Only one coordinate changes
- Others remain fixed

Why Does Gibbs Work?

- Each step satisfies detailed balance
- Hence, $\pi(x)$ is stationary

Key Insight

Gibbs sampling is a special case of Metropolis–Hastings with acceptance probability equal to 1

Limitation: Correlation

- Gibbs updates one coordinate at a time

Problem

If variables are highly correlated:

- Updates are very small
- Chain moves slowly
- Poor mixing

Question

Can we update multiple variables together?

Blocked Gibbs Sampling

Partition variables into blocks:

$$x = (x^{(1)}, x^{(2)}, \dots, x^{(k)})$$

Update

$$x^{(j)} \sim \pi(x^{(j)} \mid x^{(-j)})$$

- Update groups of variables together
- Captures dependence structure

Why Blocking Helps

- Larger moves in state space
- Reduces correlation between samples
- Improves mixing

Key Idea

Update strongly correlated variables together

Trade-off in Blocking

- Larger blocks:
  - Better mixing
  - Harder to sample from conditionals
- Smaller blocks:
  - Easier updates
  - Slower mixing

Limitation: Conditional distribution

Key Requirement

Gibbs sampling requires:

$$\pi(x_i \mid x_{-i}) \quad \text{to be known and easy to sample from}$$

- In many models:
  - Conditionals may not have closed form
  - Sampling from them may be difficult

Question

What if we cannot sample directly from $\pi(x_i \mid x_{-i})$?

Idea: Use Metropolis Instead

Approach

Replace exact conditional sampling with a Metropolis step

- At step i, instead of:

$$x_i \sim \pi(x_i \mid x_{-i})$$

- Use a proposal:

$$y_i \sim q(y_i \mid x_i)$$

- Accept or reject using MH

Metropolis-within-Gibbs

Given $X_t = x$:

1. For each coordinate i:
   - Propose:
   $$y_i \sim q(y_i \mid x_i)$$
   - Form:
   $$y = (x_1, \dots, y_i, \dots, x_d)$$
2. Accept with probability:
$$\alpha = \min\left(1,\ \frac{\pi(y)\, q(x_i \mid y_i)}{\pi(x)\, q(y_i \mid x_i)}\right)$$
3. Otherwise keep x_i

Interpretation

- Combines:
  - Gibbs (coordinate-wise updates)
  - Metropolis (accept/reject correction)

Key Idea

If exact conditional sampling is not possible, approximate it using a Metropolis step

Special Case

- If we can sample exactly:

$$q(y_i \mid x_i) = \pi(y_i \mid x_{-i})$$

Then

Acceptance probability becomes $\alpha = 1$.

Conclusion

Gibbs sampling is a special case of Metropolis-within-Gibbs

Slice Sampling: Idea

Goal: Sample from target density $\pi(x)$ (known up to normalization).

Introduce an auxiliary variable:

$$u \sim \text{Uniform}(0, \pi(x))$$

Define the slice:

$$S(u) = \{x : \pi(x) \geq u\}$$

Algorithm:

1. Sample $u \sim \text{Uniform}(0, \pi(x))$
2. Sample $x' \sim \text{Uniform}(S(u))$

Interpretation: Sampling uniformly under the graph of $\pi(x)$.

Detailed Balance and transition Kernels

We have :

$$
\begin{aligned}
K(x,x^{'}) &= \mathbb{E}_U(K(x|U,x)) \\
&= \mathbb{E}_U\left(\frac{1}{|S(U)|}\mathbf{1}(\pi(x^{'}) > U)\right) \\
&= \int \frac{1}{|S(u)|}\mathbf{1}(\pi(x^{'}) > u)\frac{1}{\pi(x)}\mathbf{1}(\pi(x) > u)du \\
\implies \pi(x)K(x,x^{'}) &= \int \frac{1}{|S(u)|}\mathbf{1}(\min(\pi(x^{'}),\pi(x)) > u)du
\end{aligned}
$$

Observe : The expression in RHS is symmetric in x and $x^{'}$. Hence we have:

$$\pi(x)K(x,x^{'}) = \pi(x^{'})K(x^{'},x)$$

Joint Distribution Representation

Define joint density:

$$p(x, u) = \mathbf{1}\{0 \le u \le \pi(x)\}$$

Then:

$$\int p(x, u)\, du = \pi(x)$$

So marginal of x is the target $\pi(x)$. Also the marginal distribution of U is:

$$\int p(x, u)\, dx = |S(u)|$$

So marginal of x is the target $\pi(x)$. Sampling (x, u) uniformly from the region under $\pi(x)$ gives samples from $\pi(x)$.

Conditional Densities

From joint:

$$p(x, u) = \mathbf{1}(0 \leq u \leq \pi(x))$$

Conditional of u given x:

$$p(u \mid x) = \frac{1}{\pi(x)} \mathbf{1}(0 \leq u \leq \pi(x))$$

Conditional of x given u:

$$p(x \mid u) = \frac{1}{|S(u)|} \mathbf{1}(x \in S(u))$$

Slice Sampling as Gibbs Sampling

The algorithm:

1. $u \sim p(u \mid x)$
2. $x' \sim p(x \mid u)$

This is exactly Gibbs sampling on (x, u).

Conclusion:

- Slice sampling is Gibbs sampling on (x, u)
- Detailed balance follows automatically

Key Insight

- Instead of sampling from $\pi(x)$ directly, we sample uniformly under its graph
- No acceptance-rejection step
- Moves are always accepted

$$\text{Complex density} \longrightarrow \text{simple uniform geometry}$$

Trace Plot

Definition

Plot of sampled values over iterations:

$$x_t \text{ vs } t$$

- Visual tool to assess convergence and mixing

Good Behavior

- Appears as a stable "horizontal band"
- Rapid fluctuations
- No visible trend

Bad Behavior

- Drift or trend $\Rightarrow$ not converged
- Long flat regions $\Rightarrow$ poor mixing

Autocorrelation Function (ACF)

Definition

Measures dependence between samples:

$$\mathrm{Corr}(X_t, X_{t+k})$$

- Indicates how quickly the chain forgets its past

Good Behavior

- Autocorrelation drops to zero quickly
- Samples are nearly independent

Bad Behavior

- Slow decay $\Rightarrow$ strong dependence
- Inefficient sampling

Multiple Chains

- Run MCMC from different initial values

Goal

Check if all chains converge to the same distribution

- Chains should mix and overlap
- Large differences $\Rightarrow$ lack of convergence

Gelman–Rubin Diagnostic ($\hat{R}$)

Setup

Run m chains from different starting points, each of length n

- Let $\bar{x}_j$ = mean of chain j
- Let $\bar{x}$ = overall mean across chains

Within-chain variance

$$W = \frac{1}{m} \sum_{j=1}^{m} s_j^2$$

where s_j^2 is the variance of chain j

Between-chain variance

$$B = \frac{n}{m-1} \sum_{j=1}^{m} (\bar{x}_j - \bar{x})^2$$

Definition and Interpretation of $\hat{R}$

Estimate of marginal variance:

$$\hat{V} = \frac{n-1}{n} W + \frac{1}{n} B$$

Potential Scale Reduction Factor

$$\hat{R} = \sqrt{\frac{\hat{V}}{W}}$$

Interpretation

- $\hat{R} \approx 1$: chains have mixed $\Rightarrow$ convergence
- $\hat{R} > 1$: chains are still different $\Rightarrow$ not converged

Intuition

If chains started far apart but now look similar, between-chain variance $\approx$ within-chain variance

Trace Plot : Example

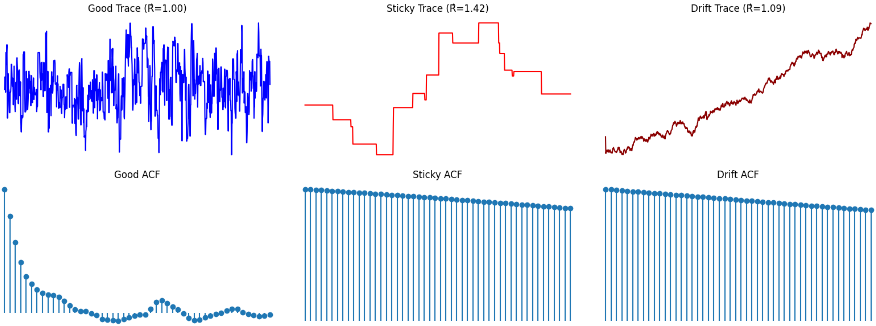

Figure: Example trace plots,ACFs and $\hat{R}$

Summary of Diagnostics

- Trace plot: checks convergence visually
- ACF: checks dependence between samples
- $\hat{R}$: checks agreement across chains

Key Insight

No single diagnostic is sufficient — use them together

Class Problems: Rejection Sampling

- **Problem 1: Optimal Proposal within a Family**
  Target: $f(x) = e^{-x^2/2}$
  Proposal: $g(x) = \text{Laplace}(0, b)$
  Find $M(b)$ and optimize over b
- **Problem 2: Mixture Proposal Design**
  Target: $0.5\,\mathcal{N}(-3, 1) + 0.5\,\mathcal{N}(3, 1)$
  Proposal: $\mathcal{N}(0, \sigma^2)$
  Find where the optimal value of M for this proposal.
  Improve using mixture proposal
- **Problem 3: Adaptive Envelope**
  Estimate M using observed max $\frac{f(x)}{g(x)}$
  What issues arise?

Class Problem: Transformation and Proposal Design

Problem 4: $f(x) = xe^{-x}, \quad x > 0$

- **(A)** Let $g(x) = e^{-x}$. Compute $\frac{f(x)}{g(x)}$ and show that no finite M exists.
- **(B)** Let $y = \log x$. Show that:

$$f_Y(y) = e^{2y} e^{-e^y}$$

- **(C)** Consider

$$g_Y(y) = \frac{\lambda}{2} e^{-\lambda|y|}$$

- **(D)** Compute $\frac{f_Y(y)}{g_Y(y)}$ and show that a finite M exists for $\lambda >= 2$.

Class Problem: Non-Reversible Chains

Problem 5: Non-Reversible Markov Chain

Consider a Markov chain on state space $\{1, 2, 3, 4\}$ with transitions:

- $1 \to 2$ with probability 1
- $2 \to 3$ with probability 1
- $3 \to 1$ with probability $\frac{1}{2}$, and $3 \to 4$ with probability $\frac{1}{2}$
- $4 \to 3$ with probability 1

- **(A)** Show that the chain is irreducible
- **(B)** Show that the chain is aperiodic
- **(c)** Find the stationary distribution π
- **(D)** Verify that detailed balance does *not* hold

Class Problems: Metropolis–Hastings

- **Problem 6: MH vs Rejection Sampling**
  Independence proposal: $q(y|x) = g(y)$
  When does MH behave like rejection sampling?
- **Problem 7: Proposal Scale Effects**
  $Y = x + \epsilon,\ \epsilon \sim \mathcal{N}(0, \sigma^2)$
  Analyze behavior as $\sigma \to 0$ and $\sigma \to \infty$
- **Problem 8: Sticky Chains**
  Construct proposal with high self-transition probability
  Effect on convergence?
- **Problem 9: Acceptance Geometry**
  Interpret $\alpha(x, y) = \min(1, r)$
  Relation to optimization vs sampling